

Smart PPE for Workers in Hostile environments: Use of Electroencephalography (EEG) and Near-Infrared Spectroscopy (NIRS) for Situational Awareness classification

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Dear Sir,

I submit this thesis entitled “Smart PPE for Workers in Hostile environments: Use of Electroencephalography (EEG) and Near-Infrared Spectroscopy (NIRS) for Situational Awareness classification”, based on MXEN4000 Mechatronic Engineering Research Project 1 and MXEN4004 Mechatronic Engineering Research Project 2, undertaken by me as part-requirement for the degree of B.Eng. in Mechatronic Engineering.

Yours faithfully,

Ankush Dugar
20172345

Acknowledgments

This thesis has been a challenging but rewarding journey and I am deeply thankful to everyone who helped make it possible. I would like to take a moment to express my appreciation to those who supported me along the way.

First and foremost, I want to sincerely thank Anjula De Silva and Jhon Castaneda Gonzalez. Their quick responses and constant support made a huge difference throughout my project. They ensured I had all the equipment I needed and were instrumental in collecting the data that was vital for this thesis.

A big thank you also goes to Adil Khokar, Tony Comerford, David Geraghty and Sarah Yu for giving me the amazing opportunity to collaborate with Mineral Resources Ltd. on this project. Their efforts in organising this partnership were invaluable and made the experience even more special.

I am also incredibly grateful to my colleagues and friends from Mineral Resources Ltd.: Lauren Rothnie, Samantha Taylor, Rehaab Syed, Benjamin Portlock, Mathieu Widjaja, Johann Lim, Nathan Broadbent, Ahmed Mohammed, and Eric Lau. Your guidance, friendship and support were crucial throughout this journey and I truly appreciate the great company and help you provided.

Lastly, I would like to thank my supervisor Dr. Hanieh Bakhshayesh for giving me the opportunity to work on such an interesting and rewarding topic. Your support and feedback throughout the project were invaluable and I am deeply grateful for your guidance.

This thesis would not have been possible without the help of all these wonderful people. I am truly thankful for the support I received, both personally and professionally and I will always be grateful for your contributions.

Abstract

In high-risk industries like mining, construction, and aviation, maintaining situational awareness (SA) is crucial for operational safety and preventing accidents. This thesis explores the foundational development of a smart helmet which could monitor SA in real-time aiming to enhance worker safety in hostile environments. Specifically, it investigates Electroencephalography (EEG) and Near-Infrared Spectroscopy (NIRS) to classify different SA levels. EEG tracks electrical brain activity while NIRS measures blood oxygenation and these combined provide a more comprehensive assessment of cognitive states. The study involved a data collection experiment where participants performed situational awareness tests, based on the SAGAT framework, while brain activity was recorded using EEG and NIRS. After pre-processing the data to remove noise and artefacts, key features were extracted from the signals including power spectral density, statistical measures as well as Independent Component Analysis (ICA) and Principal Component Analysis (PCA) components. These features were used to train four machine learning models: k-Nearest Neighbours (KNN), XGBoost, Multi-Layer Perceptron (MLP), and a Transformer model. The Transformer model achieved the highest accuracy at 60.3% outperforming the other models by capturing intricate relationships in the data with its self-attention mechanism. However, the study also faced challenges particularly in EEG signal quality and the limitations of using a single participant, which affected the generalisability of the models. Future work should focus on improving EEG signal quality, collecting data from multiple participants and using raw time-series data for better temporal modelling with the Transformer, potentially increasing classification accuracy. This thesis demonstrates the potential of integrating EEG and NIRS into smart PPE for real-time SA monitoring and lays the groundwork for future advancements in proactive worker safety technologies.

Chapter 1, 2 and 4.1 of this thesis were previously submitted as part of the Progress Report assessment for the MCEN4005 Mechanical Engineering Research Project 1 unit.

Table of Contents

Acknowledgments.....	i
Abstract	ii
Table of Contents	iii
List of Tables.....	vii
List of Figures	vii
1 Introduction	1
1.1 Ethical, Professional and Cultural Considerations	1
1.2 Scope	2
1.3 Significance of the Study	2
2 Background Information and Literature Review	3
2.1 The importance of Situational Awareness.....	3
2.2 Approaches to measure Situational Awareness.....	3
2.3 Brain activity	4
2.4 Electroencephalography and Near-Infrared Spectroscopy	5
2.5 Previous similar studies.....	6
2.6 Methods for Situational Awareness data collection	7
3 Aims	9
4 Data Collection Procedure	10
4.1 Materials	10
4.1.1 OpenBCI Cyton Biosensing Board (EEG).....	10
4.1.2 Artinis OctaMon mini (fNIRS)	11
4.2 Software.....	12
4.2.1 SAGAT Based Situational Awareness Psychological Test	12
4.2.2 OpenBCI GUI	16
4.2.3 OxySoft (by Artinis)	17
4.3 Experimental Procedure	18

4.3.1	Physical Setup	18
4.3.2	Software setup	22
4.3.3	Process	25
5	Project Code	26
6	Data Pre-processing	27
6.1	EEG data.....	27
6.1.1	Loading into python	27
6.1.2	Pre-processing	28
6.2	fNIRS data	29
6.2.1	Loading into python	29
6.2.2	Pre-processing	30
6.3	PEBL SA Test data.....	31
6.3.1	Loading	31
6.3.2	Categorising data.....	32
7	Feature Extraction	34
7.1	Isolating Data Samples	34
7.2	EEG Feature Extraction.....	35
7.2.1	Power Spectral Density	35
7.2.2	Signal Statistical Features	36
7.2.3	Principal Component Analysis.....	36
7.2.4	Independent Component Analysis	37
7.3	fNIRS Feature Extraction.....	37
7.3.1	Signal Statistical Features	37
7.3.2	Independent Component Analysis	38
7.4	Feature Selection	38
7.4.1	Recursive Feature Elimination (RFE).....	38
7.4.2	Consideration of Other Methods.....	39

7.4.3	Top 10 Features.....	39
8	Model Development.....	41
8.1	Model 1 – k-Nearest Neighbours (KNN).....	41
8.1.1	Background	41
8.1.2	Implementation	42
8.2	Model 2 – XGBoost (Boosted Trees).....	43
8.2.1	Background	43
8.2.2	Implementation	44
8.3	Model 3 – Multi-layer Perceptron	45
8.3.1	Background	45
8.3.2	Implementation	45
8.4	Model 4 – Transformer.....	47
8.4.1	Background	47
8.4.2	Implementation	47
9	Results and Discussion.....	50
9.1	Model Performance	50
9.2	Discussion	50
9.2.1	Overall Model Performance	50
9.2.2	Performance Across Metrics	51
9.2.3	Comparison to Literature	52
9.2.4	Lessons from the Use of Extracted Features.....	53
10	Conclusions from the work	54
11	Future work	56
11.1	Data Collection and Generalisability.....	56
11.2	EEG Signal Quality and Impedance.....	56
11.3	Model Improvement: Use of Raw Time-Series Data	57
11.4	Thresholding and SA Test Design.....	57

11.5	Cross-Model Comparison and Hybrid Approaches	58
12	References	59
13	Appendices.....	66
13.1	Appendix A: Multi-Input CNN with windowing technique	66
13.2	Appendix B: Data Pre-Processing Code	67

List of Tables

Table 1. SA Test probes summary.	16
Table 2. Code dependencies.	26
Table 3. SA Level respective DataFrame columns.	32
Table 4. Top 10 features identified by RFE.	40
Table 5. Model performance results.	50

List of Figures

Fig. 1 Block diagram representing data collection experiment procedure.	10
Fig. 2 OpenBCI Cyton Biosensing Board (left side) and USB dongle (right side) [31].	11
Fig. 3 OpenBCI Gold Cup EEG Electrodes.....	11
Fig. 4 fNIRS Artinis OctaMon mini.	12
Fig. 5 Flow diagram of SAGAT based Situational Awareness test.....	13
Fig. 6 PEBL SA test grid of animals.....	14
Fig. 7 Test 1: Perception (Level 1)	14
Fig. 8 Test 2: Comprehension (Level 2)	15
Fig. 9 Test 3: Projection (Level 3).....	15
Fig. 10 OpenBCI GUI Example widget layout.....	17
Fig. 11 OxySoft interface example.	18
Fig. 12 Artinis fNIRS device - 8 channel setup.	18
Fig. 13 EEG device setup.....	19
Fig. 14 USB dongles for EEG and fNIRS devices.	19
Fig. 15 Nuprep Skin Prep Gel.....	20
Fig. 16 Ten20 conductive paste.	20
Fig. 17 Images of participant wearing the EEG and fNIRS devices.....	21
Fig. 18 8-channel EEG electrode placement [37]......	21
Fig. 19 EEG impedances for the data collection session.	22
Fig. 20 OxySoft new measurement setup.	23
Fig. 21 Connecting to the Cyton board with the OpenBCI GUI.	24
Fig. 22 PEBL launcher.....	24
Fig. 23 Snippet of a raw EEG .txt file.....	27
Fig. 24 EEG data before pre-processing.	28

Fig. 25 EEG Data after outlier removal and pre-processing.....	29
Fig. 26: Snippet of raw fNIRS .txt file.	29
Fig. 27 Snippet of Legend and other information provided in the data file.....	30
Fig. 28 fNIRS data channel before pre-processing.....	31
Fig. 29 fNIRS data channel after pre-processing.....	31
Fig. 30: Simple graphic to visualise the KNN algorithm (k=3).....	42
Fig. 31 Architecture of MLP implemented in this thesis.....	46
Fig. 32 Architecture of Transformer model implemented in this thesis.....	49
Fig. 33 Structure of the multi-cnn deep neural network [54].	66

1 Introduction

Personal Protective Equipment (PPE) has undergone continuous development across various industries over the past decades, serving as the first line of defence against occupational hazards faced by workers. However, ensuring worker safety still remains the highest priority which requires further advancements and specialisation of PPE for diverse high-risk tasks [1]. The advancements that have occurred in sensor technology, wearable devices and machine learning presents a unique opportunity to develop the next generation of PPE – smart PPE. A potential device that could be part of smart PPE is a smart helmet.

Situational awareness (SA) refers to the ability to perceive, comprehend and predict critical aspects of an environment to make informed decisions and execute tasks safely and efficiently. In high-risk industries such as construction, mining and aviation, maintaining high situational awareness is essential for both operational success and the prevention of accidents and injuries. A smart helmet capable of real-time monitoring and feedback on an individual's awareness levels could prove invaluable for both workers and supervisors. This research project delves into the foundational technologies required for such a product with specific focus on the development of a situational awareness detection system utilising Electroencephalography (EEG) and Near-Infrared Spectroscopy (NIRS).

1.1 Ethical, Professional and Cultural Considerations

The integration of wearable technologies into the workplace also raises several ethical, professional and cultural issues. Ethically, monitoring workers' cognitive states introduces concerns about privacy and worker autonomy. Employers need to carefully consider how this data is collected, stored and used. Ensuring transparency and obtaining informed consent from workers is critical to maintaining trust. Additionally, safeguards must be implemented to prevent the misuse of cognitive data, such as penalising workers for temporary lapses in attention rather than addressing the underlying causes (e.g., fatigue, stress).

From a professional standpoint, the adoption of wearable situational awareness systems has the potential to enhance worker safety and improve job performance.

However, it is important to establish clear guidelines and regulations around the implementation of these systems to prevent any misuse that might place undue stress on employees. Workers must be trained not only to use these technologies but also to understand the limitations of the systems. Furthermore, the integration of AI-based systems into safety-critical environments should be governed by ethical AI principles to ensure fairness, transparency and accountability.

In terms of cultural considerations, the implementation of these technologies must consider differences in workplace culture across industries and regions. In some cultures, the introduction of surveillance technologies might be viewed as intrusive or overly controlling even if it is for safety purposes. Engaging with workers and stakeholders during the design and implementation stages can help foster a sense of inclusion and mitigate resistance to these technologies. Workplace culture and labour laws also vary between regions requiring customised approaches to ensure that these systems respect local norms and regulations.

1.2 Scope

The scope of this project is to explore the foundational technology required for a smart helmet that can detect someone's awareness. Specifically, the project will determine the accuracy to which situational awareness can be classified via the combinational use of EEG and NIRS data.

1.3 Significance of the Study

The significance of this study lies in its potential to transform workplace safety across various high-risk industries such as construction, mining and transportation. By integrating advanced brain monitoring technologies into PPE, this research aims to address the limitations of current safety measures which often rely on passive protection and post-incident analysis. Smart PPE can actively monitor and respond to the cognitive states of workers which could potentially prevent accidents before they occur. Furthermore, the development of such technology can set new standards for occupational safety which could influence regulations and policies worldwide.

2 Background Information and Literature Review

2.1 The importance of Situational Awareness

Situational awareness is a person's perception, cognitive thinking and resulting action within the environment they are present in [2]. A loss of situational awareness can occur when people are fatigued because of high physical and mental loads. People working in complex, hostile and unsafe environments are required to have a high level of awareness as small errors in their tasks can result in equipment damage, injuries and even fatalities. As per the Australian Transport Accident Commission, 20% of all fatal crashes in Victoria, Australia on the road involve driver fatigue [3]. Another example is the incident at the Upper Big Branch mine in West Virginia in April 2010, where an explosion within the coal mine occurred due to inadequate ventilation and poor safety practices, resulting in the death of 29 miners and injuring several other personnel [4]. It was determined from the investigations that a lack of situational awareness among mine workers and management and insufficient monitoring of methane gas levels resulted in the outcome. Hence, wearable Personal Protective Equipment (PPE) that can provide real-time awareness monitoring could allow for preventative measures to be taken and reduce risks associated with working in hostile and dangerous environments.

2.2 Approaches to measure Situational Awareness

A widely accepted model of situational awareness is Endsley's three level model based on an information processing approach [5]. The model states that awareness can be divided into three levels, where achieving level 3 correlates to a high level of awareness where a person can act to the best of their ability to complete their task [6]. Level 1 corresponds to the ability to perceive different variables within the environment, Level 2 corresponds to appropriately comprehending the situation based on the perceived variables, and Level 3 corresponds to the ability to predict the future state given their perception and comprehension of the current state [6].

Many subjective approaches towards measuring awareness exist and involve having participants perform tasks and answering questionnaires. A few types of approaches include freeze probe techniques, real-time probe techniques, observer rating techniques, self-rating techniques and performance measures [5]. Many previous

studies use the results of these subjective approaches to determine scores for each of the three levels of situational awareness [7]. However, these methods can be inaccurate because the participants can provide incorrect responses as a person's perception of their own awareness can be affected by multiple variables.

Objective approaches can be more accurate in determining a subject's situational awareness as they leverage physiological and behavioural measures. Physiological sensors, such as electromyography (EMG) to measure muscle tension and electrodermal activity (EDA) to track sweat gland activity, can provide insights into a subject's stress and fatigue levels [8] [9]. Additionally, electrooculography (EOG) to monitor eye movements, electrocardiography (ECG) to assess heart rate variability, and photoplethysmography (PPG) to measure blood volume changes can offer further information about a person's physiological state in relation to alertness and sleepiness [10] [11].

2.3 Brain activity

A physiological signal which is promising for the comprehension of a person's mental state is brain activity. Brain activity plays a crucial role in all levels of situational awareness, from perception to comprehension to projection. Different brain regions are responsible for processing sensory information, integrating it with past experiences and ultimately guiding decision-making and actions. Some brain regions include:

- Perception: The primary visual, auditory, and somatosensory cortices are responsible for receiving and interpreting sensory information from the environment [12].
- Attention: The frontal lobe plays a key role in directing attention to relevant stimuli and filtering out distractions, critical for maintaining focus in dynamic situations [13].
- Working Memory: Fronto-parietal brain regions, including the prefrontal, cingulate, and parietal cortices are activated when information is retained and manipulated to complete tasks successfully [14].
- Decision-making: The prefrontal cortex and hippocampus are heavily involved in decision-making [15].

The specific relationship between brain waves and awareness are pointed out below.

- Delta Waves (0 – 4Hz): Associated with deep sleep, unconsciousness and restorative processes. High delta activity can indicate a state of rest or unconsciousness [16].
- Theta Waves (3 – 7Hz): Involved in memory, learning, and mental workload; theta activity can fluctuate depending on the complexity of the situation and the cognitive demands placed on the individual [17].
- Alpha Waves (8 – 12Hz): Associated with relaxation and alertness; high alpha activity might indicate a state of decreased focus that could hinder situational awareness [18].
- Beta Waves (13 – 30Hz): Linked to active concentration and focussed attention; increased beta activity might be observed during tasks requiring high situational awareness [18].
- Gamma Waves (30 – 45Hz): Linked to high-level cognitive functions such as problem-solving, attention, and perception. Increased gamma activity is often associated with heightened awareness, information processing and the integration of sensory information during complex tasks [19].

Brain activity can provide valuable physiological information that can be insightful towards the situational awareness of a person. Two prominent techniques for non-invasive measurement of brain activity are Electroencephalography (EEG) and Near-Infrared Spectroscopy (NIRS).

2.4 Electroencephalography and Near-Infrared Spectroscopy

EEG measures the electrical activity of the brain generated by the synchronised firing of large populations of neurons [20]. Electrodes placed on the scalp detect tiny voltage fluctuations resulting from these neuronal currents. Data can be provided in both time (frequency) and voltage (amplitude) domains, hence allowing for the analysis of different types of brain waves [21]. EEG offers high temporal resolution, meaning it can capture rapid changes in brain activity with millisecond precision [20]. This makes it well-suited for studying the dynamic nature of situational awareness, where brain activity can fluctuate quickly as the environment changes or new information is received. However, EEG signals can be susceptible to electrical noise from sources like muscle activity and eye movements [20]. Additionally, EEG has limited spatial resolution, making it difficult to pinpoint the exact location of brain activity [20].

NIRS measures changes in blood oxygenation levels by transmitting sufficient photons in the near infrared region (700-1300nm) through organs [22]. Specifically, the information provided in the infrared spectrum can determine the reduction of cytochrome a,a3 (Cyt.Ox.) by measuring the change in absorption band of Cyt.Ox., which provides information on the oxygen availability at the intracellular level [22]. Specifically for measurements at the brain, it transmits near-infrared light through the scalp and skull, and the amount of light absorbed by tissues is measured by detectors. Since active brain regions require more oxygenated blood, changes in blood oxygenation levels can be indirectly linked to brain activity [23]. NIRS offers good spatial resolution (up to 1cm), meaning it can provide information about activity in specific brain regions [24]. Additionally, NIRS is portable and less susceptible to movement artefacts compared to EEG, making it potentially more suitable for real-world applications in dynamic environments where workers might be moving around. However, NIRS has a poorer temporal resolution (typically between 1-10Hz) compared to EEG which essentially means NIRS signals are slower than EEG signals [24]. This indicates that they may not capture the most rapid changes in brain activity.

2.5 Previous similar studies

There are a few studies that relate to the correlation of EEG with situational awareness. Kästle, et al. [25] use a 32-channel Nautilus wireless EEG headset with dry electrodes to measure the brain activity of participants at a rate of 250Hz. In this study, the PEBL situational awareness test framework, which is based off the Situational Awareness Global Assessment Technique (SAGAT), is used for the program that classifies the data into either high or low situational awareness for each level of awareness [25]. The data has been labelled using position and angular error values from the PEBL data collection, ensuring that the labelling has been done objectively. Kastle, et al. [25] use Independent Component Analysis (ICA) and Principal Component Analysis (PCA) to then decompose the EEG signal and extract individual features. The classification methods used in the study consist of Random Forrest and Boosted Tree, which obtained the highest of test accuracies of 66.8% and 65.8% respectively.

Yeo, et al. [26] use a 14-channel Emotiv semi-wet wireless EEG headset with a sampling rate of 128Hz. This study uses the Situation-Present Assessment Method (SPAM) which is a one of the commonly used methods for measuring situational

awareness [26]. It utilises this method by having participants maintain aircraft separation and control air traffic flow, whilst responding to SPAM probes to measure their workload and situational awareness [26]. Linear regression, extreme learning machine and support vector regression for regression were used to correlate the EEG data to the situational awareness data and a correlation factor of 0.73, 0.68 and 0.72 was found respectively [26].

Fewer studies relating to the correlation of data obtained from NIRS and situational awareness exist. Bracken, et al. [27] use the fNIRS Pioneer, a single channel system, and fNIRS Explorer, a 2-channel system. The driving hazard awareness task, which is implemented throughout many countries' procedure for obtaining a driver's license, was used to label the NIRS data for the training of the models [27]. The features that were used as inputs to the logistic regression model were signed oxyhemoglobin concentration amplitude, unsigned oxyhemoglobin concentration amplitude, fNIRS signal direction, oxyhemoglobin concentration variance and the derivative of oxyhemoglobin concentration [27]. A mean accuracy of 69% for predicting level 3 situational awareness in individual participants was achieved [27].

A particular study combines the 2 technologies of EEG and NIRS. Nguyen, et al. [28] predict driver drowsiness using the blinking rate, eye closure, heart rate, alpha and beta band power signals which are obtained from the combination of EEG and NIRS. The data was collected from a simulated driving task, with subjects experienced in both awake and drowsy states [28]. The combination of EEG/NIRS outperformed the individual NIRS and EEG methods, achieving a mean accuracy of 79.2% which is 8.7% higher in comparison to EEG alone and 5.5% higher in comparison to NIRS alone [28].

2.6 Methods for Situational Awareness data collection

The current study aims to assess situational awareness with the objective method of brain activity measured by EEG and NIRS technologies. However, establishing a link between these brain activity signals and awareness levels requires training data that has been categorised with appropriate scores for awareness. Prior studies investigating awareness through neurophysiological measures have employed various data

collection methodologies, including simulated driving assessments [28], driving hazard perception tests [27], and maintaining aircraft separation tasks [26].

Two primary approaches exist for measuring situational awareness: freeze-probe and real-time probe techniques. Freeze-probe techniques, such as the widely used Situation Awareness Global Assessment Technique (SAGAT) initially developed for pilots [7], involve interrupting the task at random intervals and presenting participants with situational queries [5]. Participant responses are then compared to a pre-defined ground truth of the scenario, yielding a final situational awareness score. Real-time probe techniques, on the other hand, administer situational queries either before or during the task itself [5]. Metrics like response time and accuracy are used to assess awareness levels. The Situation Present Assessment Method (SPAM) represents the most common real-time probe technique [5].

Amongst various situational awareness measurement methods, SAGAT emerges as a particularly effective metric [29]. This preference stems from SAGAT's established validity and reliability, as demonstrated by Endsley [30] in their comprehensive review of theoretical underpinnings for SA measurement [30]. Additionally, Salmon, et al. [5] supported the effectiveness of SAGAT within stable task environments where elements, states, and properties can be accurately defined. This finding is supported by the positive correlation observed between SAGAT scores and performance in a controlled experimental environment utilising Brunel University's command wall [5].

3 Aims

Previous research, particularly the combined EEG-NIRS approach used by Nguyen et al. [28], has shown promise in improving classification accuracy. This study aims to replicate this success by applying a similar combination for classifying levels of situational awareness. EEG and NIRS can address each other's limitations: while EEG offers high temporal resolution for capturing fast brain activity, its spatial resolution (identifying specific brain regions) is limited. Conversely, NIRS excels at spatial resolution but struggles with capturing rapid changes. Combining them can provide a more comprehensive picture.

This thesis focuses on the following key aims:

- Designing a Data Collection Experiment
 - To classify situational awareness levels based on EEG and NIRS readings, we need data in both high and low awareness states. This experiment will be designed to induce these contrasting states.
- Developing/Procuring Data Collection Software
 - Software is required that effectively induces different awareness states in the participant when they wear the EEG and NIRS devices. This might involve a simple game or task based on the SAGAT approach.
- Developing Data Labelling code
 - The data obtained from the brain activity devices and the SAGAT based game will have to be linked together so that the brain activity data is labelled based on the scores of the game.
- Data Processing
 - Pre-processing pipelines will be implemented using the obtained data to establish a valid pipeline for training our model on “meaningful” data inputs.
- Model Building and Training
 - Different models will be trained on the data and the performance of each of these will be recorded.
- Review and Conclusion
 - Compare and contrast the models and conclude on the outcomes of the research project.

4 Data Collection Procedure

In order to develop a model that can classify a person's situational awareness based on their brain activity, labelled data is crucial. This labelled data will establish a correlation between the EEG and NIRS signals and different levels of situational awareness. To achieve this, a data collection experiment has been designed that simultaneously monitors brain activity (EEG and NIRS) while measuring situational awareness through a separate method.

The high-level procedure for the data collection experiment is depicted in the block diagram in Figure 1.

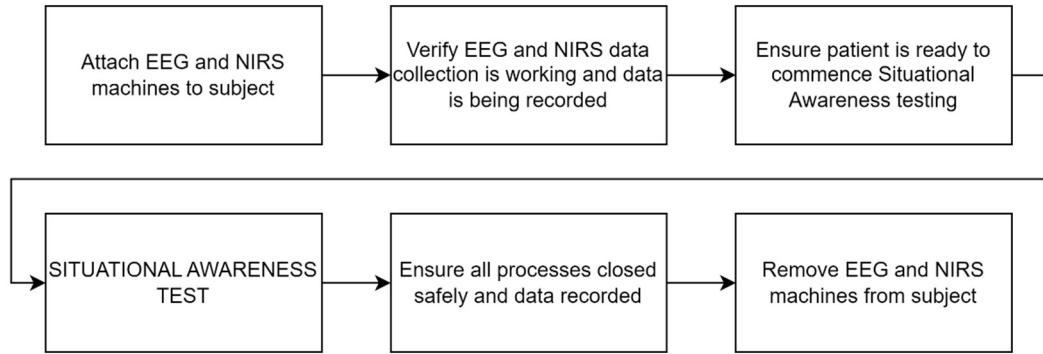


Fig. 1 Block diagram representing data collection experiment procedure.

4.1 Materials

4.1.1 OpenBCI Cyton Biosensing Board (EEG)

The experiment utilises the OpenBCI Cyton Biosensing Board (Figure 2) for EEG signal acquisition.

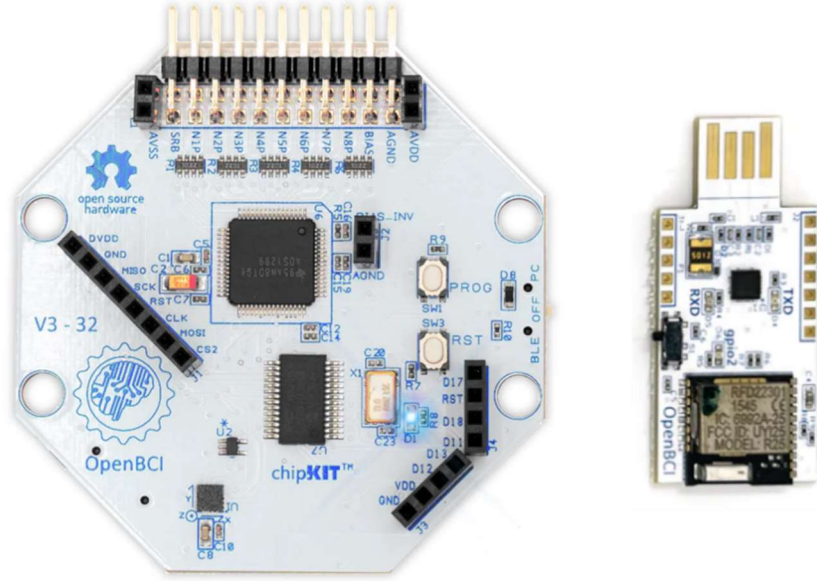


Fig. 2 *OpenBCI Cyton Biosensing Board (left side) and USB dongle (right side) [31].*

The board provides a sampling rate of 250 samples per second when using 8 channels and 125 samples per second with 16 channels [32]. Data will be transmitted wirelessly via Bluetooth to a computer using a dedicated USB dongle which is also shown in Figure 2. EEG electrodes, shown in Figure 3, connect to the designated pin headers on the board. The board is also battery-powered to minimise electrical noise contamination, particularly at the common 60Hz frequency [32].



Fig. 3 *OpenBCI Gold Cup EEG Electrodes.*

4.1.2 Artinis OctaMon mini (fNIRS)

For functional near-infrared spectroscopy (fNIRS) data acquisition, the portable OctaMon mini device from Artinis is employed which can be seen in Figure 4. This

device offers 8 channels and provides coverage across the entire prefrontal cortex [33]. Notably, the OctaMon mini measures changes in oxyhemoglobin, deoxyhemoglobin, and total hemoglobin concentrations [33].



Fig. 4 fNIRS Artinis OctaMon mini.

4.2 Software

4.2.1 SAGAT Based Situational Awareness Psychological Test

In Figure 1, there is a block called “Situational Awareness Test”. This section details the implementation of the situational awareness (SA) assessment procedure used to label the EEG and NIRS data. The assessment is based on the SAGAT framework and a block diagram of the assessment is shown in Figure 5 below.

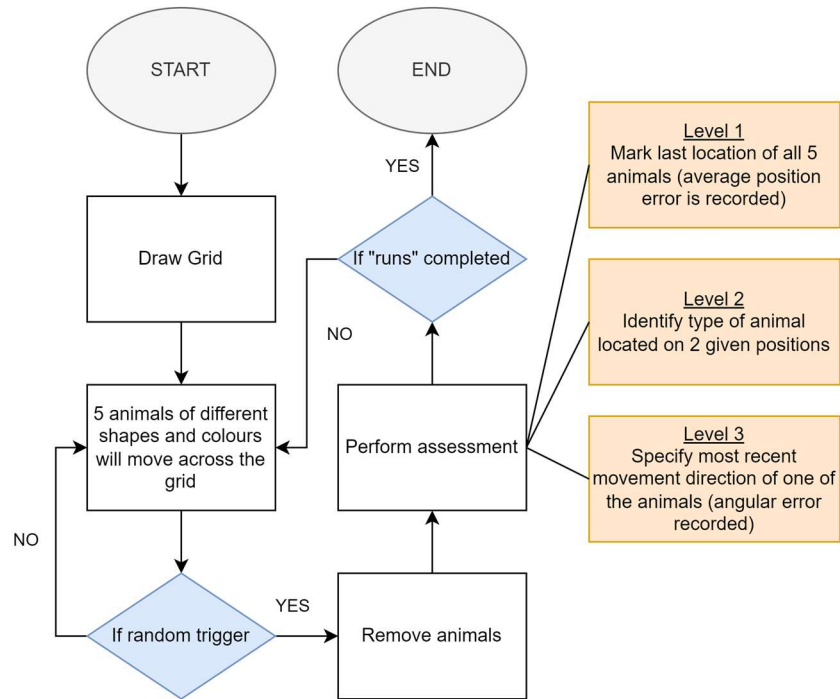


Fig. 5 Flow diagram of SAGAT based Situational Awareness test.

The assessment utilises the PEBL (Psychology Experiment Building Language) Situational Awareness test. PEBL is an open-source software which is capable of letting users design, run and share behavioural tests [34]. The software also comes with multiple common tests for usage in the research community within the PEBL test battery [34]. Within this test battery, the SA assessment this study will take advantage of is present.

In the SA assessment, participants interact with a visual environment displayed on a computer screen (see Figure 6). This environment features 5 moving animals (2 lizards and 3 insects) on a 2D grid. The animals move autonomously within the grid boundaries with insects respawning in random locations upon being “eaten” by lizards.

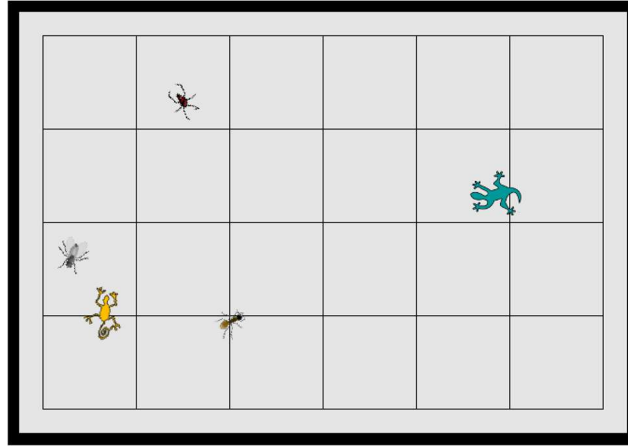


Fig. 6 PEBL SA test grid of animals.

During the assessment, the simulation is paused at random intervals. Upon each pause, participants are presented with a series of questions designed to evaluate their SA score at that specific moment. 3 distinct test types will be employed, corresponding to 3 levels of the SAGAT framework.

Test 1 assesses the participant's ability to perceive the current state of the environment. Participants are asked to identify the most recent locations of all 5 animals (Figure 7). The average error between their selections and the true animal positions is used as a performance metric. This test assesses SA level 1.

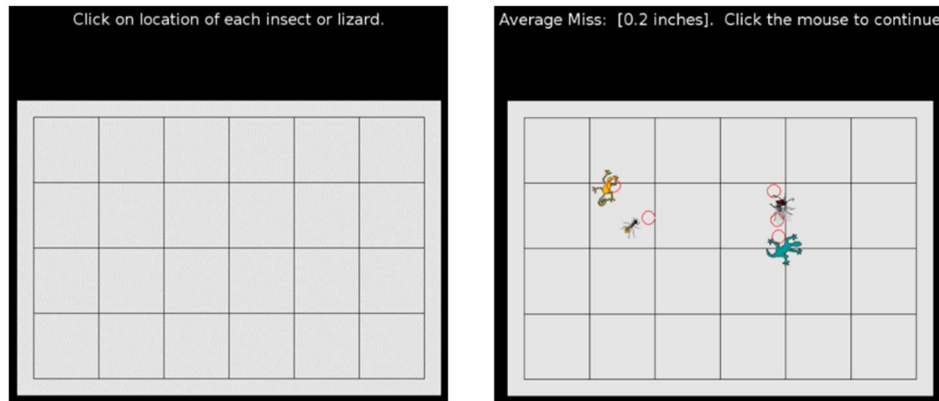


Fig. 7 Test 1: Perception (Level 1)

Test 2 evaluates the participant's understanding of the current situation. Participants are presented with the environment and tasked with correctly identifying the type of animal at 2 designated locations marked by larger circles (see Figure 8). A binary score (correct/incorrect) for each identified animal is used as the performance metric.



Fig. 8 Test 2: Comprehension (Level 2)

Test 3 assesses the participant's ability to predict future states. Participants are presented with the location and type of a specific animal and asked to identify the direction of its movement (see Figure 9). The angular error between the predicted and actual direction of movement is used as the performance metric.

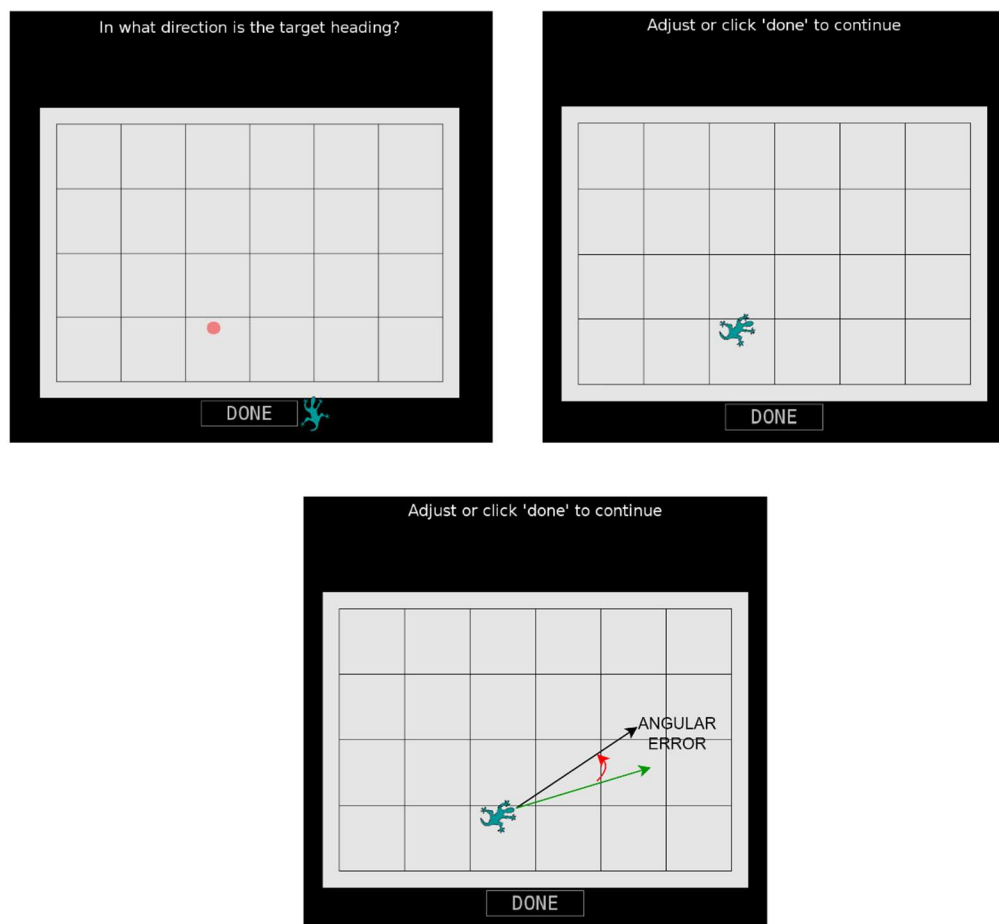


Fig. 9 Test 3: Projection (Level 3)

By collecting performance data from these 3 tests at various points throughout the simulation, the corresponding EEG and NIRS brain activity data can be labelled. This labelled data is then used to build and train classifiers. Table 1 below show a summary of the different probes.

Table 1. SA Test probes summary.

Probe Description	SA Level	Performance Metric
Identify all animal positions.	1	Average position error
Determine which animal was at which position (2 animals)	2	2 binary scores, 1 for each animal
Identify direction of particular animal's movement.	3	Angular error

4.2.2 OpenBCI GUI

To interface with the Cyton board, the first step is to connect the USB dongle to a PC. The OpenBCI GUI, a free software tool provided by OpenBCI, is then be used to read and manage the data transmitted by the board. This software provides the ability to visualise, record and stream the EEG data collected by the Cyton board.

The OpenBCI GUI is utilised to establish a connection with the Cyton board and save the data in a .txt format for further analysis. Additionally, the GUI offers valuable visualisations that facilitate the experimental setup. For example, it provides an impedance check for each electrode, ensuring that the connections are functioning properly before data collection begins. Figure 10 below shows some of the widgets available in the GUI.



Fig. 10 OpenBCI GUI Example widget layout.

4.2.3 OxySoft (by Artinis)

To interface with the fNIRS OctaMon device, a USB dongle is connected to the PC. OxySoft is a proprietary recording and analysis software developed by Artinis specifically for their devices [35]. This software is used to read and manage the data transmitted by the fNIRS device and provides a comprehensive interface for real-time monitoring. An example of the interface is shown in Figure 11 below.

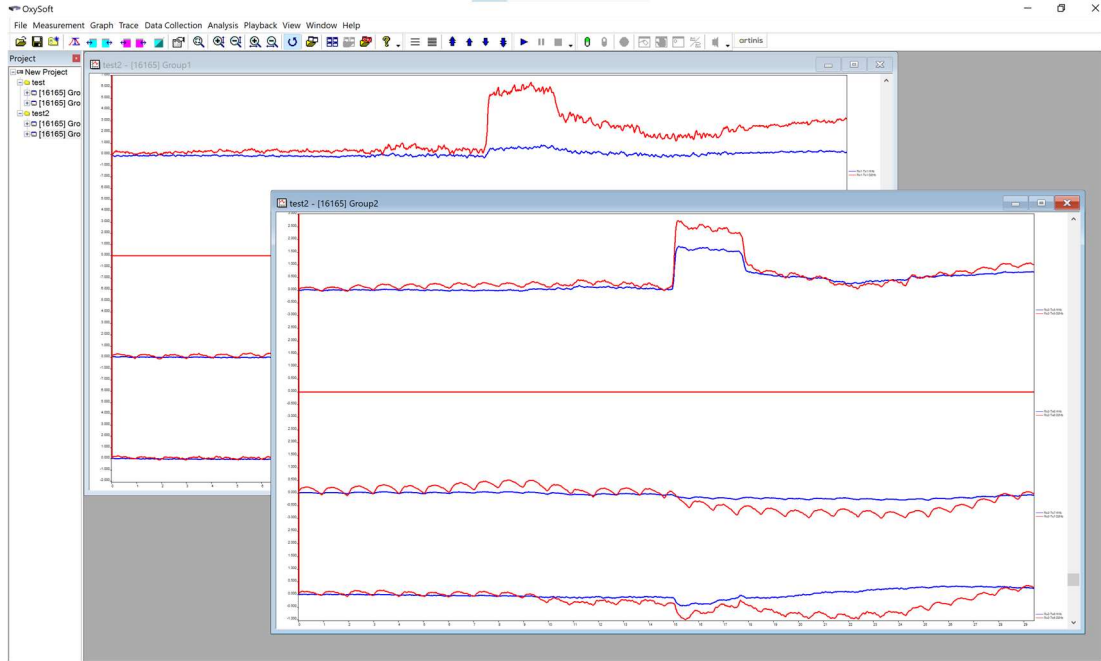


Fig. 11 OxySoft interface example.

4.3 Experimental Procedure

4.3.1 Physical Setup

The setup process began with configuring the fNIRS device to use an 8-channel arrangement, with each receiver paired with a pattern of 4 transmitters. Figure 12 shows the headband setup for this 8-channel configuration.

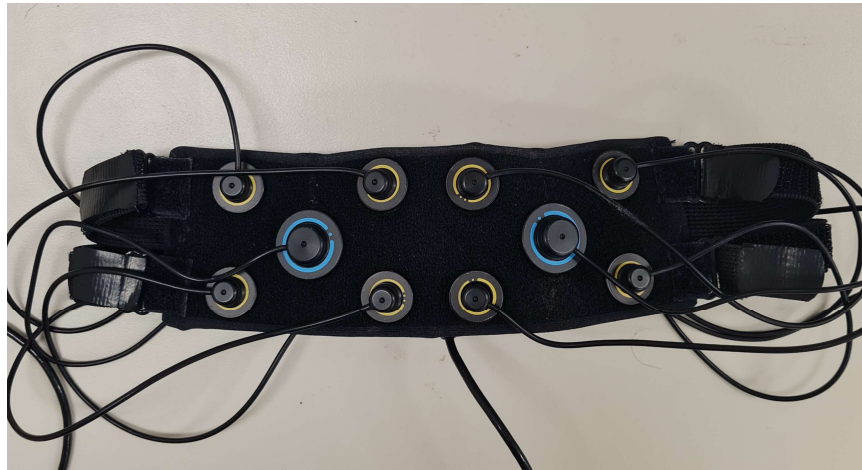


Fig. 12 Artinis fNIRS device - 8 channel setup.

Next, the EEG device was configured for 8-channel use by connecting the battery and electrodes in the appropriate pattern. The specific setup of the EEG device is shown in Figure 13.

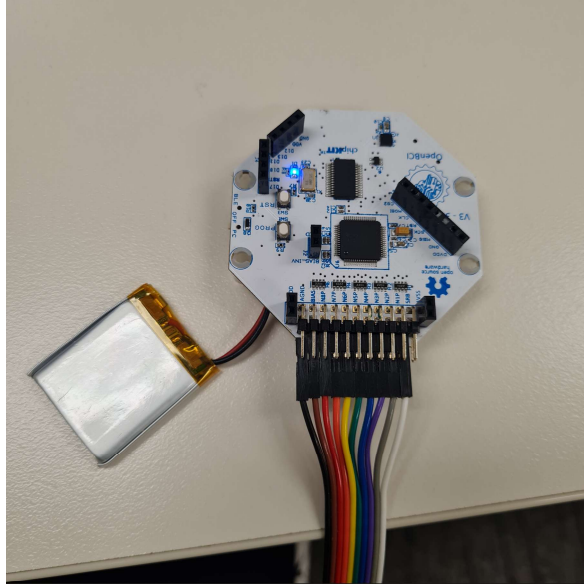


Fig. 13 EEG device setup.

The USB dongles for both the EEG and fNIRS devices were then connected to the same laptop, as shown in Figure 14. Using a single laptop for data collection from both devices is crucial to maintain consistency in the timing of samples and the recording of timestamps.

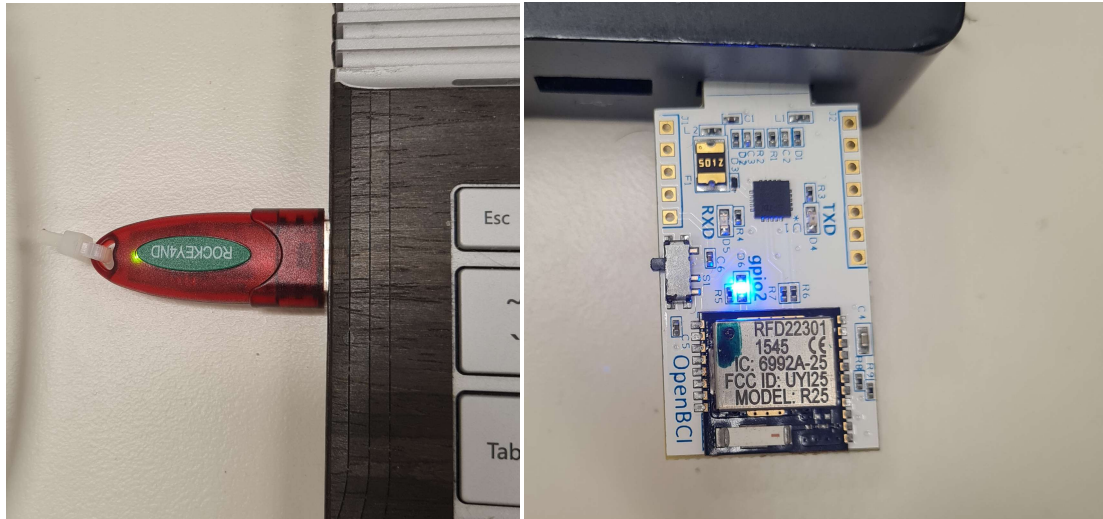


Fig. 14 USB dongles for EEG and fNIRS devices.

Following this, skin preparation gel was applied to the areas of the scalp where the EEG electrodes were to be placed. The skin was lightly scrubbed until slight redness appeared to reduce skin impedance to minimise noise at the electrode interface [36]. Figure 15 shows the Nuprep Skin Prep Gel used for this purpose.



Fig. 15 Nuprep Skin Prep Gel.

Once the skin was prepped, conductive paste was applied to the electrodes to further decrease impedance. The Ten20 conductive paste used in this experiment is shown in Figure 16.



Fig. 16 Ten20 conductive paste.

The EEG electrodes were then placed into the appropriate locations on the scalp and the fNIRS headband was worn by the participant. Any loose cables were secured to the participant with masking tape to prevent movement artifacts. Figure 17 below shows the participant wearing both devices.



Fig. 17 Images of participant wearing the EEG and fNIRS devices.

Figure 18 below illustrates the 10-20 electrode placement system used for the 8-channel EEG setup, indicating the various brain regions: frontal lobe (F), temporal lobe (T), central lobe (C), parietal lobe (P), and occipital lobe (O) [37].

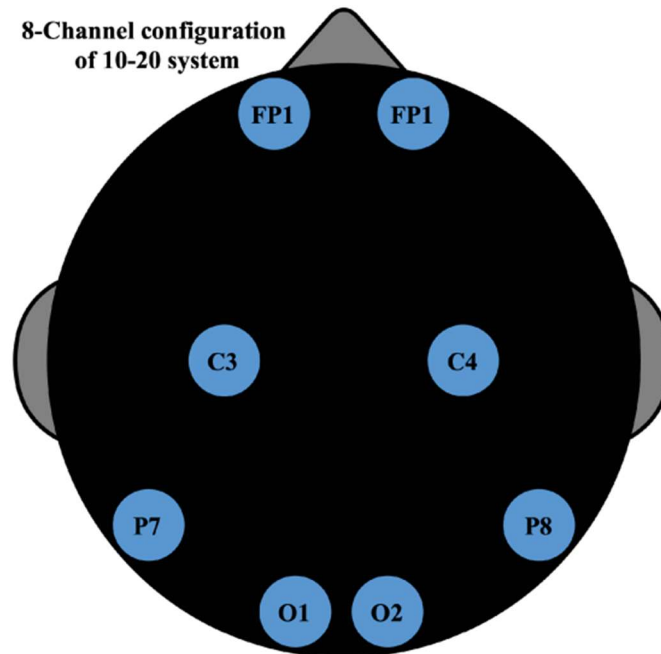


Fig. 18 8-channel EEG electrode placement [37].

As each electrode was positioned, an impedance check was run using the OpenBCI GUI to ensure that the impedance levels were within an acceptable range. Figure 19 shows the actual impedances recorded during the data collection session.

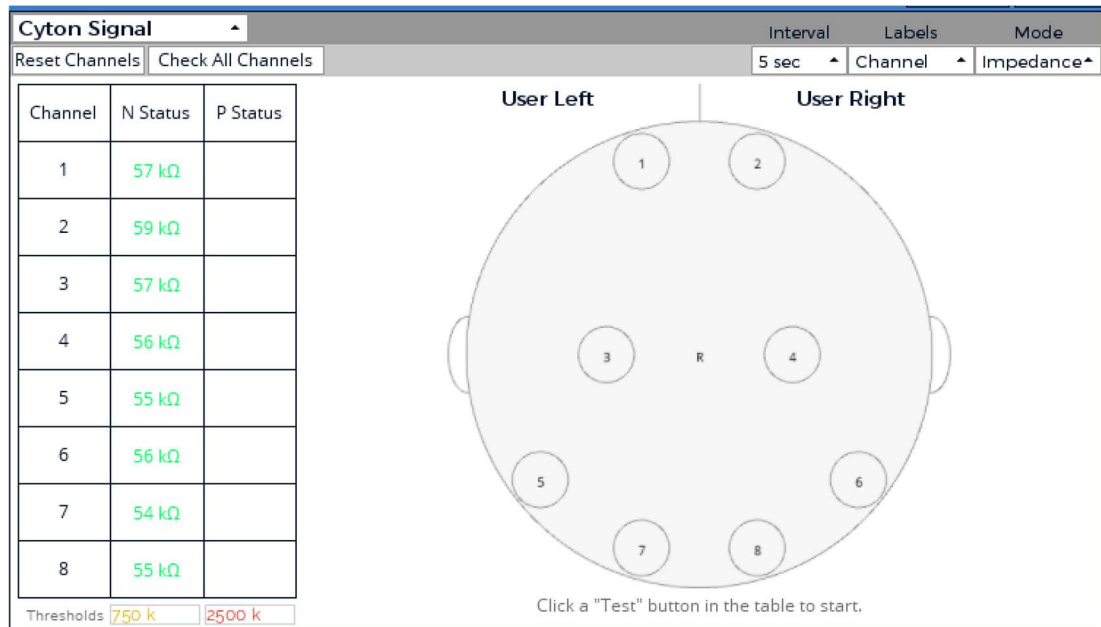


Fig. 19 EEG impedances for the data collection session.

Upon reviewing the data, it was noted that the impedance values recorded may have been too high potentially affecting the quality of the EEG data. In future experiments, maintaining an impedance of less than 20 kilo-ohms would be ideal as this is closer to the standards used in medical settings.

4.3.2 Software setup

For data collection with the OxySoft software, a new measurement has to be created, the device chosen and importantly the physical configuration of the transmitters and receivers must be provided to the software. Figure 20 shows some important parts of the setup process.

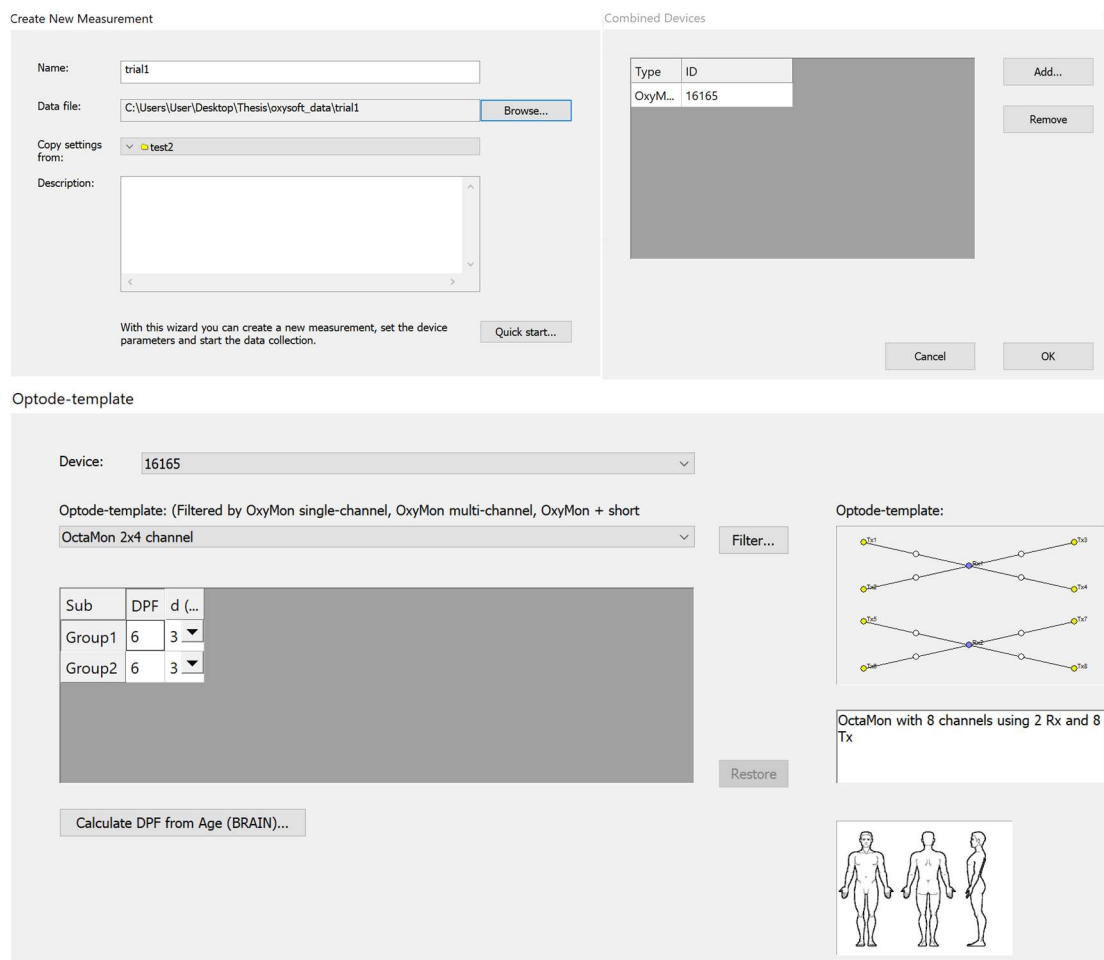


Fig. 20 OxySoft new measurement setup.

For setting up data collection with the OpenBCI GUI, the process is much simpler with the only requirement being the Cyton board has to be identified and connected. Figure 21 shows the process of connecting to the board with the GUI.

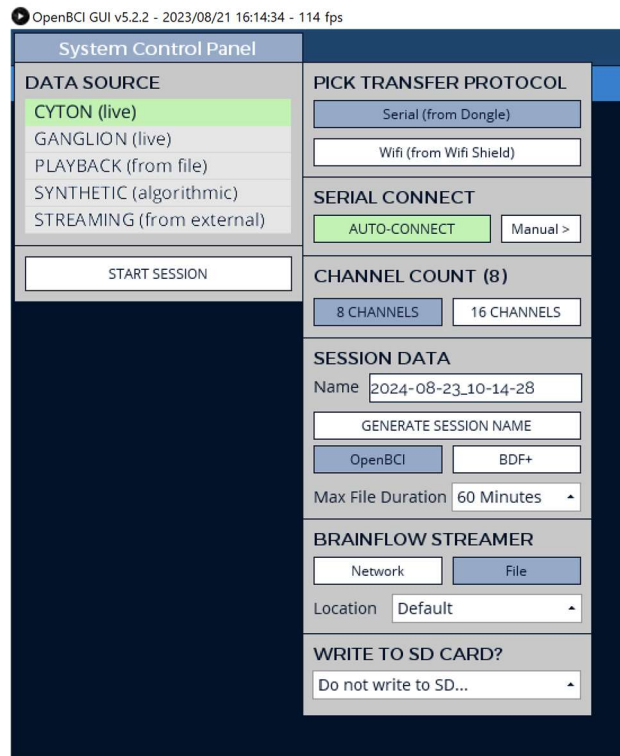


Fig. 21 Connecting to the Cyton board with the OpenBCI GUI.

In order to start the SA test, the PEBL launcher is used where some settings can be configured if desired and the test can be started. Figure 22 shows the PEBL launcher.

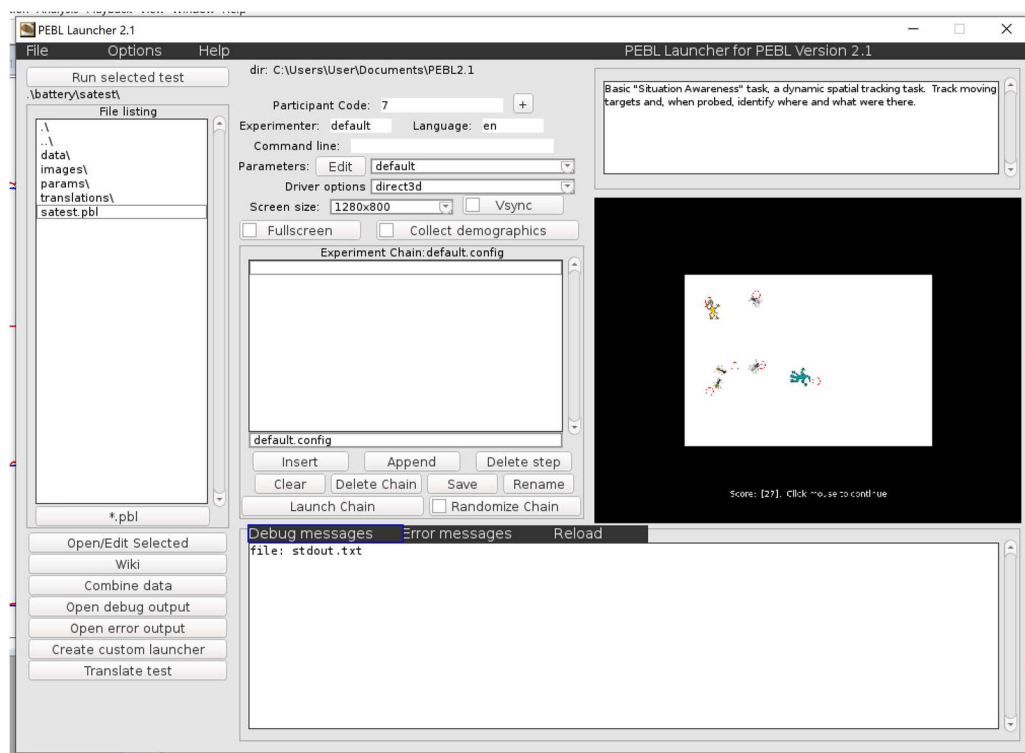


Fig. 22 PEBL launcher.

4.3.3 Process

Once the setup was completed, the EEG and fNIRS data collection sessions were started. Afterwards, the SA Test was started and the participant followed the prompts of the test. Each trial involved initially 3 runs of each SA level probe to make the participant comfortable with the way the test works. Then, 4 phases occurred: phase 1 involved just SA level 1 probes, phase 2 involved just SA level 2 probes, phase 3 involved just SA level 3 probes and phase 4 involved all SA level probes in a random occurrence. In total, 90 probes were performed for each trial, 30 for each SA level.

The participant completed 7 trials but due to some technical issues, only 5 trials have valid data. This means that data for 450 probes were collected in total, 150 for each SA level. Including the $9 \times 5 = 45$ practice probes performed, the dataset consists of a total of 495 samples.

5 Project Code

The code for this thesis was written in Python version 3.10.14. Table 2 the packages that are dependencies for the code in this thesis.

Table 2. Code dependencies.

Package Name	Version
DateTime	5.5
imblearn	0.0
matplotlib	3.9.2
numpy	1.26.4
pandas	2.2.2
scikit-learn	1.5.2
scipy	1.14.1
seaborn	0.13.2
tensorflow	2.17.0
tqdm	4.66.5
xgboost	2.1.1

The data and code for this thesis can be requested for from the supervisor of this project (Dr. Hanieh Bakhshayesh) or a relevant Curtin University/Mineral Resources Ltd. contact. It is not available publicly due to Intellectual Property agreements.

6 Data Pre-processing

6.1 EEG data

6.1.1 Loading into python

The OpenBCI GUI allowed a recorded session's data containing the readings of each electrode and the associated timestamps to be placed into a .txt file which was well formatted. This format allows the data to be easily imported into Python for further analysis. A snippet of the raw EEG data from such a .txt file is shown in Figure 23.

```
%OpenBCI Raw EXG Data
%Number of channels = 8
%Sample Rate = 250 Hz
%Board = OpenBCI_GUI$BoardCytonSerial
Sample Index, EXG Channel 0, EXG Channel 1, EXG Channel 2, EXG Channel 3, EXG Channel 4, EXG Channel 5, EXG Channel 6, EXG Channel 7, Accel
0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.7243793585934484E9, 0.0, 2
1.0, 160724.35357860968, 55782.555733031724, 70683.27911892881, 79409.35545079177, 57028.28550675935, 70154.81682477199, 74843.9669423064, 1
2.0, 163810.59304602066, 58850.55617696717, 73772.49136835235, 82484.82448396976, 60104.15376474307, 73183.94758509965, 77923.6349684751, 1
3.0, 165924.44222264795, 60973.033126954215, 75891.52614969328, 84526.42971592305, 62222.22742107242, 75306.93862520918, 80042.82624039963, 1
4.0, 162654.62787802552, 57737.19343390387, 72618.38139514702, 81399.4653781145, 58972.596701693146, 72124.05021477345, 76782.15375925944, 1
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6.0, 160787.85799120166, 55762.640328722046, 70662.31318262973, 79390.06589532684, 57009.97942805045, 70133.806184984, 74821.10110772862, 10
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70.0, 159867.6335653822, 54961.84438012176, 69836.8856116397, 78615.53324646155, 56184.90948497171, 69335.78185269616, 73992.29842332583, 10
71.0, 160704.34293882168, 55771.648081737534, 70666.69412454298, 79370.731636373, 57005.17380299256, 70132.44272857222, 74814.12736345857, 1
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78.0, 165953.29832473974, 61013.91446756297, 75930.39583330105, 84532.64350088162, 62252.26816562035, 75329.17861094221, 80070.27418259074, 1
79.0, 162724.0076928148, 57819.73842617732, 72699.18295135295, 81449.33211199428, 59042.26708916034, 72187.59622425989, 76852.87467871603, 1
80.0, 159867.6335653822, 54961.84438012176, 69836.8856116397, 78615.53324646155, 56184.90948497171, 69335.78185269616, 73992.29842332583, 10
81.0, 160704.34293882168, 55771.648081737534, 70666.69412454298, 79370.731636373, 57005.17380299256, 70132.44272857222, 74814.12736345857, 1
82.0, 163765.97896408784, 58816.44741492837, 73735.07454813416, 82341.21141924994, 60061.216063644424, 73134.97491299808, 77874.05882764564, 1
83.0, 165953.29832473974, 61013.91446756297, 75930.39583330105, 84532.64350088162, 62252.26816562035, 75329.17861094221, 80070.27418259074, 1
84.0, 162724.0076928148, 57819.73842617732, 72699.18295135295, 81449.33211199428, 59042.26708916034, 72187.59622425989, 76852.87467871603, 1
85.0, 159867
```

`readEEGData()` and `readAllEEGData()` handle the process of loading the EEG data into a DataFrame for further analysis.

6.1.2 Pre-processing

Once the data was loaded into a dataframe, several pre-processing steps were applied to the EEG signals to clean and prepare the data. The common steps included applying both a notch filter and a band-pass filter.

A notch filter was applied at 50 Hz to remove power line noise as 50 Hz is the mains power frequency in Australia where this experiment was conducted. To further reduce noise, a band-pass filter was used to attenuate frequencies above 45 Hz and below 0.5 Hz. Frequencies below 0.5 Hz are typically dominated by slow drift artefacts which is why this lower threshold was selected [38]. The upper threshold of 45 Hz was chosen because gamma brain waves, which are relevant for the analysis, fall within the 30 to 45 Hz range [25]. The final pre-processing step involved removing outliers which simply involved replacing data points that were 4 standard deviations away with the median.

Figure 24 and Figure 25 below show a channel of the EEG data before and after pre-processing respectively. Note that the plot contains the signal data for all trials which is why there are some “gaps” present.

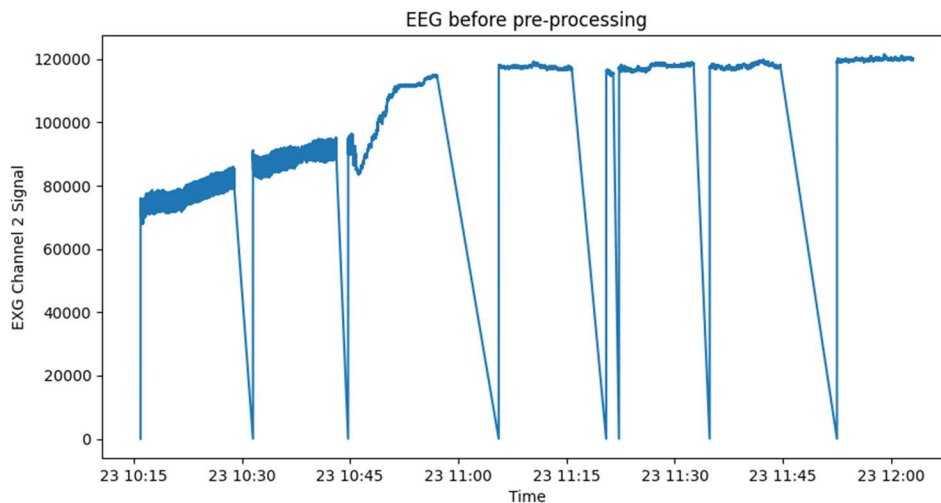


Fig. 24 EEG data before pre-processing.

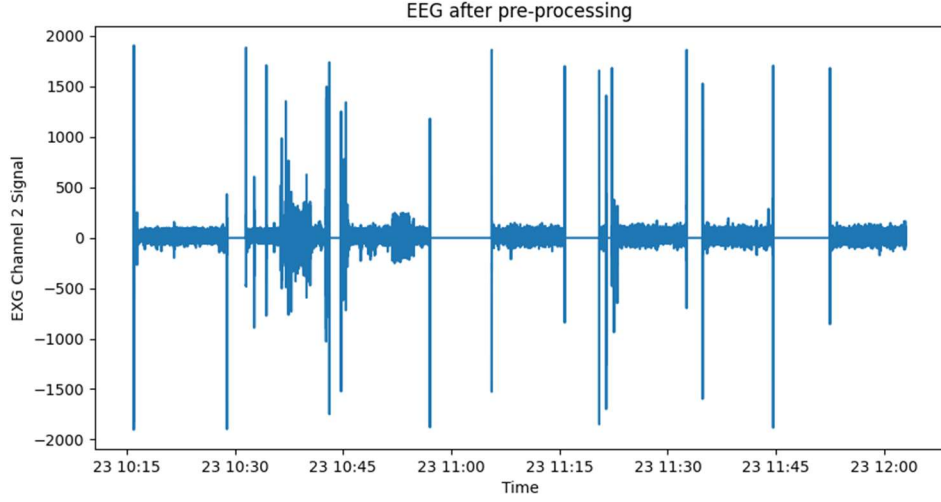


Fig. 25 EEG Data after outlier removal and pre-processing.

The pre-processing steps were performed in the function *EEGData()* within the code.

6.2 fNIRS data

6.2.1 Loading into python

The OxySoft software offers an option to export data in the more commonly used .txt format, as opposed to the less conventional and software specific .oxy4 format. The well-organised structure of the .txt file allows it to be easily imported into Python. A snippet of the raw fNIRS data is shown in Figure 26.

```

1      2      3      4      5      6      7      8      9      10      11      12      13      14      15      16      17      18
0 53.0435 53.9175 -0.0004 0.0000 71.7426 61.5140 64.4355 61.3587 80.9328 59.3911 -0.0004 0.0000 94.0458 72.3200 91.8262 65.2507
1 -0.2815 0.0670 -0.0004 0.0000 0.0587 0.0498 0.1269 0.0249 0.2520 0.0220 -0.0004 0.0000 0.2320 0.0364 0.1967 0.0279
2 -0.0860 -0.0398 -0.0004 0.0000 0.0715 0.0309 0.0651 0.0403 0.2039 0.0324 -0.0004 0.0000 0.2039 0.0324 0.1822 0.0190
3 -0.1180 0.0074 -0.0004 0.0000 0.0779 0.0214 0.0779 0.0214 0.1750 0.0145 -0.0004 0.0000 0.1758 0.0284 0.1613 0.0195
4 0.0320 -0.0472 -0.0004 0.0000 0.0779 0.0214 0.0234 0.0412 0.1541 0.0150 -0.0004 0.0000 0.1541 0.0150 0.1469 0.0105
5 -0.2239 -0.0180 -0.0004 0.0000 0.0153 0.0229 -0.0192 0.0283 0.1397 0.0060 -0.0004 0.0000 0.1260 0.0110 0.1188 0.0065
6 -0.3419 -0.0106 -0.0004 0.0000 -0.0473 0.0244 -0.1291 0.0541 0.1397 0.0060 -0.0004 0.0000 0.0979 0.0070 0.1115 0.0020
7 -0.3739 0.0366 -0.0004 0.0000 -0.0689 0.0109 -0.1227 0.0446 0.0979 0.0070 -0.0004 0.0000 0.0698 0.0030 0.0971 -0.0069
8 -0.3739 0.0366 -0.0004 0.0000 -0.0906 -0.0025 -0.2069 0.0326 0.1107 -0.0119 -0.0004 0.0000 0.0417 -0.0010 0.0762 -0.0064
9 -0.3739 0.0366 -0.0004 0.0000 -0.1251 0.0029 -0.1941 0.0138 0.0626 -0.0015 -0.0004 0.0000 0.0273 -0.0099 0.0554 -0.0060
10 -0.2495 0.0198 -0.0004 0.0000 -0.0906 -0.0025 -0.1388 0.0078 0.0418 -0.0010 -0.0004 0.0000 0.0064 -0.0095 0.0481 -0.0104
11 -0.0860 -0.0398 -0.0004 0.0000 -0.0626 0.0015 -0.0545 0.0198 0.0273 -0.0099 -0.0004 0.0000 0.0064 -0.0095 0.0554 -0.0060
12 -0.1180 0.0074 -0.0004 0.0000 -0.0064 0.0095 0.0651 0.0403 0.0690 -0.0109 -0.0004 0.0000 0.0281 0.0040 0.0834 -0.0020
13 -0.2495 0.0198 -0.0004 0.0000 0.0498 0.0175 0.2111 0.0369 0.0835 -0.0020 -0.0004 0.0000 0.0770 0.0075 0.1260 0.0110
14 -0.2495 0.0198 -0.0004 0.0000 0.0996 0.0349 0.3444 0.0523 0.0915 0.0164 -0.0004 0.0000 0.1268 0.0249 0.1750 0.0145
15 -0.2815 0.0670 -0.0004 0.0000 0.0932 0.0443 0.3869 0.0653 0.1750 0.0145 -0.0004 0.0000 0.1902 0.0374 0.2311 0.0225
16 -0.3995 0.0744 -0.0004 0.0000 0.1494 0.0523 0.4222 0.0738 0.2312 0.0225 -0.0004 0.0000 0.2673 0.0449 0.2937 0.0210
17 -0.2495 0.0198 -0.0004 0.0000 0.2401 0.0548 0.5458 0.0430 0.2593 0.0265 -0.0004 0.0000 0.3098 0.0578 0.3427 0.0245
18 -0.2495 0.0198 -0.0004 0.0000 0.2618 0.0683 0.5330 0.0619 0.2946 0.0349 -0.0004 0.0000 0.3452 0.0663 0.3571 0.0335
19 -0.0000 0.0000 -0.0004 0.0000 0.3180 0.0762 0.6300 0.0550 0.3010 0.0255 -0.0004 0.0000 0.3733 0.0703 0.3780 0.0330
20 -0.0256 0.0378 -0.0004 0.0000 0.3460 0.0802 0.6854 0.0491 0.3219 0.0250 -0.0004 0.0000 0.4078 0.0648 0.3716 0.0424
21 -0.0256 0.0378 -0.0004 0.0000 0.3805 0.0748 0.6854 0.0491 0.3427 0.0245 -0.0004 0.0000 0.4222 0.0738 0.3852 0.0375
22 -0.0256 0.0378 -0.0004 0.0000 0.3805 0.0748 0.6790 0.0585 0.3155 0.0344 -0.0004 0.0000 0.4431 0.0733 0.3716 0.0424
23 -0.0256 0.0378 -0.0004 0.0000 0.3805 0.0748 0.6662 0.0774 0.3299 0.0434 -0.0004 0.0000 0.4503 0.0778 0.3852 0.0375
24 -0.0576 0.0850 -0.0004 0.0000 0.3869 0.0654 0.6790 0.0585 0.3716 0.0424 -0.0004 0.0000 0.4639 0.0728 0.3861 0.0514
25 -0.1755 0.0924 -0.0004 0.0000 0.3588 0.0614 0.5491 0.0987 0.3716 0.0424 -0.0004 0.0000 0.4712 0.0773 0.3997 0.0464
26 -0.0256 0.0378 -0.0004 0.0000 0.3869 0.0654 0.5491 0.0987 0.3444 0.0524 -0.0004 0.0000 0.4712 0.0773 0.3997 0.0464

```

Fig. 26: Snippet of raw fNIRS .txt file.

In addition to the data, the file includes useful metadata about the data collection process and details about the dataset itself, as illustrated in Figure 27.

```

OxySoft export of: C:\Users\User\Desktop\Thesis\oxysoft_data\trial3
Export date: 2024/08/23 10:59:20
Start of measurement: 2024-08-23 10:44:02.908

Datafile sample rate: 50.00 Hz
Datafile duration: 787.760 s
Datafile total number of samples: 39388

Description:
Optode-template: OctaMon 2x4 channel
Optode distance (mm): 35.00 35.00
Optode gradients (mm): - -
DPF: 6.00 6.00

Device id: 16165
# Receivers: 2
# Light sources: 16
# ADC: 1

Export sample rate: 50.00 Hz
Selected time span: 0 787.760
Selected time span (sample numbers): 0 39388

Legend:
Column Trace (Measurement)
1 (Sample number)
2 Rx1-Tx1 O2Hb (trial3)
3 Rx1-Tx1 HHb (trial3)
4 Rx1-Tx2 O2Hb (trial3)
5 Rx1-Tx2 HHb (trial3)
6 Rx1-Tx3 O2Hb (trial3)
7 Rx1-Tx3 HHb (trial3)
8 Rx1-Tx4 O2Hb (trial3)
9 Rx1-Tx4 HHb (trial3)
10 Rx2-Tx5 O2Hb (trial3)
11 Rx2-Tx5 HHb (trial3)
12 Rx2-Tx6 O2Hb (trial3)
13 Rx2-Tx6 HHb (trial3)
14 Rx2-Tx7 O2Hb (trial3)
15 Rx2-Tx7 HHb (trial3)
16 Rx2-Tx8 O2Hb (trial3)
17 Rx2-Tx8 HHb (trial3)
18 (Event)

```

Fig. 27 Snippet of Legend and other information provided in the data file.

Similar to the EEG data, the fNIRS data was consolidated into one pandas.DataFrame using the functions `readNIRSData()` and `readAllNIRSData()`. The final DataFrame contained the following columns: Sample Number, Rx1-Tx1 O2Hb, Rx1-Tx1 HHb, Rx1-Tx2 O2Hb, Rx1-Tx2 HHb, Rx1-Tx3 O2Hb, Rx1-Tx3 HHb, Rx1-Tx4 O2Hb, Rx1-Tx4 HHb, Rx2-Tx5 O2Hb, Rx2-Tx5 HHb, Rx2-Tx6 O2Hb, Rx2-Tx6 HHb, Rx2-Tx7 O2Hb, Rx2-Tx7 HHb, Rx2-Tx8 O2Hb, Rx2-Tx8 HHb.

6.2.2 Pre-processing

The data exported from OxySoft in .txt format has already undergone some initial processing. Specifically, the values do not appear to represent raw optical density (OD) but rather the concentrations of deoxygenated hemoglobin (HHb) and oxygenated hemoglobin (O2Hb). According to Artinis [39], this is because the Modified Beer-Lambert Law (MBLL) has already been applied to the data.

As a result, the main remaining pre-processing steps involve applying a low-pass filter and removing outliers. A low-pass filter is necessary to attenuate high-frequency noise. For the fNIRS data in this thesis, a low-pass filter with a cutoff frequency of 0.7 Hz was applied. This filter is essential for mitigating physiological and instrumental noise, such as interference from extraneous light and physiological artefacts like cardiac pulsation and respiration [40].

Figure 28 and Figure 29 below show a channel of the fNIRS data before and after pre-processing respectively.

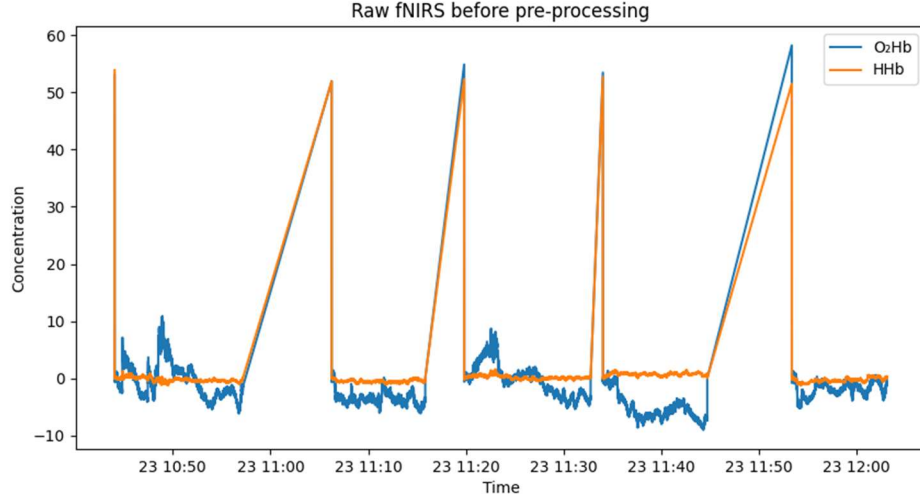


Fig. 28 *fNIRS data channel before pre-processing.*

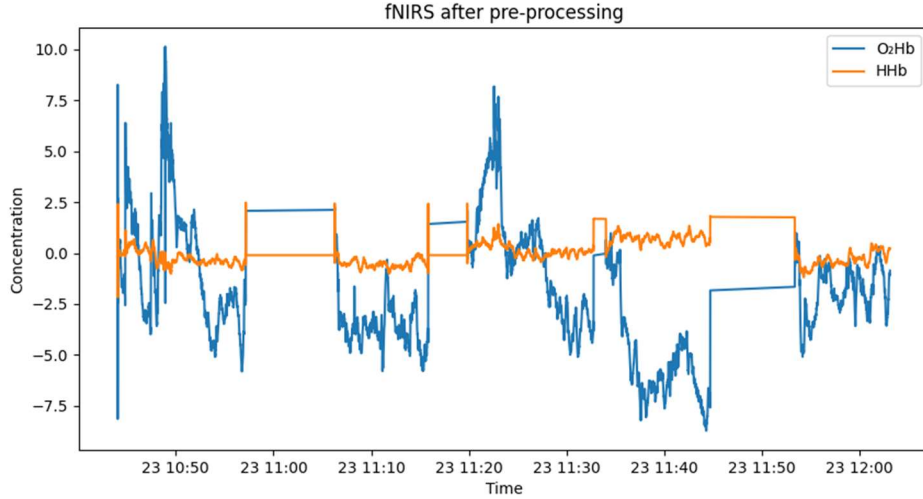


Fig. 29 *fNIRS data channel after pre-processing.*

The pre-processing steps were performed in the function *NIRSData()* within the code.

6.3 PEBL SA Test data

6.3.1 Loading

The data generated by the PEBL SA test was poorly formatted and required more extensive cleaning and reformatting. Due to the lack of documentation on the data generated by the PEBL SA test, the data was cleaned by investigating the source code of the test written in the PEBL language (with file extension .pbl). Through this analysis, several useful functions were developed to convert the raw data into three

pandas DataFrames – one for each level of SA. These functions may prove valuable for future researchers who plan to use the same SA test.

The first step involved determining the start time of the test, as the timestamps within the data file were represented by integers denoting the number of milliseconds since the test began. The *findSAStartTime()* function was created to extract the start time as a datetime object. With the start time identified, the function *readSATestData()* was designed to load the raw data into three separate DataFrames, each corresponding to a different SA level. Each DataFrame contains columns specific to its respective SA level, as outlined in Table 3 below.

Table 3. *SA Level respective DataFrame columns.*

SA Level	Columns
Level 1	click_number, resp.x, resp.y, click_time, avg_error
Level 2	click_number, target_id, target.x, click_time, result
Level 3	target_id, target_1.x, target_1.y, radius, angle, click_time, angle_diff

These functions streamline the data cleaning process, making it easier to work with and analyse the SA test data.

6.3.2 Categorising data

The interpretation of the SA data is performed similarly to [25]. Samples of each type of assessment (level 1, level 2 and level 3) are labelled into either the *high* or *low* SA class. These categories have been defined based on the requirements of real-world applications such as operating heavy machinery or other autonomous systems. The argument is that a human who does not have high SA cannot safely operate an autonomous system [25].

Due to the limitations of time for the completion of this thesis, it is difficult to obtain data from a wide variety of people. Hence, threshold values for level 1 and level 3 that determine high and low SA are taken from [25]. In [25], the same PEBL software is used and so these threshold values correspond directly for the implementation required in this thesis. Using the data of 32 subjects, the probability distribution of the level 1

and 3 error values was modelled using a 2-component Gaussian mixture model [25]. The intersection of the 2 Gaussian distributions was used to define the threshold for high or low SA [25].

For level 1, the threshold is 0.49 hpx. So error values above this are considered low SA. For level 3, the threshold is 37.43° so error values above this are considered low SA. For the level 2 test, both answers must be correct to classify as high SA [25].

The function *clasifySATestData()* creates a new column in each SA level DataFrame which is given a value of “False” for low SA and “True” for high SA based on the thresholds. Finally, the function *SAData()* combines all of the separate trials into 3 large DataFrames for the 3 different SA levels. This function also has the option of *combine = True*, which merges the 3 DataFrames for each SA level into 1.

7 Feature Extraction

7.1 Isolating Data Samples

The brain activity data and the SA test data are timestamped using the same computer. This allows the brain activity data to be mapped onto the results of the SA tests. The maximal offset between the data points of the EEG and SA test will be equal to the temporal resolution of the OpenBCI Cyton board. The board is capable of 250 samples per second which is a temporal resolution of 4ms. Similarly, the maximal offset between the fNIRS data and the SA test will be 20ms as the OctaMon is collected data at 50 samples per second. With the assumption that the participant's SA is independent between runs, the outcome of each test can be taken as an individual sample.

Hence, the brain activity data in the time t before the moment when a probe occurred and only when the animals were moving on the screen was isolated. These isolated EEG and fNIRS signals were correlated to the associated label provided by the SA test result. These individual signals can now have features extracted to be used as inputs for classification models.

Originally, windowing was applied to the EEG and fNIRS signals to ensure uniform input sizes and to increase the number of samples in total. The windowing was done such that each EEG window was 250 data points and each fNIRS window was 50 data points, both corresponding to a 1 second duration. All windows that were within the time period when the participant was performing an SA test were used and labelled based on the probe results for that particular test.

However, this windowing approach did not provide great results and is flawed due to the nature of the data collection process. For instance, 1 second of brain activity of the individual at the beginning of a probe in comparison to at the end of the probe will have many differences due to external factors. Hence, labelling these to be the same SA result may not be accurate. Instead, the entire brain activity during the period of the test will most likely hold the information that provides appropriate insight into the level of SA of the individual. The windowing approach has been used in other literature but it is not appropriate for this particular experiment. Appendix A contains some detail on work that was performed using this method which involved the implementation of a multi-input convolutional neural network (CNN) model.

Therefore, feature extraction on the entire individual sample is necessary to create consistent input dimensions while also capturing higher-level patterns in the data that provide a more comprehensive representation of the participant's brain activity during the SA test.

7.2 EEG Feature Extraction

7.2.1 Power Spectral Density

One of the most commonly used methods for extracting features from EEG signals is through the analysis of Power Spectral Density (PSD). PSD represents how the power of a signal is distributed across different frequency components [41] and is particularly useful in identifying patterns within specific EEG frequency bands, such as delta (0 – 4 Hz), theta (4 – 8 Hz), alpha (8 – 12 Hz), beta (12 – 30 Hz), and gamma (30 – 45 Hz) [42]. These bands are known to correspond to various cognitive processes and states. In this work, the PSD of EEG signals was calculated using both the Fast Fourier Transform (FFT) and periodogram methods, with the Welch method applied to estimate the PSD for each EEG channel.

The extracted features included the average power in each EEG band for every channel. This was calculated by integrating the power within the corresponding frequency ranges. These features help capture the rhythmic activity within the brain which is essential for understanding the different levels of mental engagement during the SA tests. The use of both FFT and periodogram methods provided a comprehensive approach to analysing the frequency domain of the signals as an FFT provides information of how much of each frequency is present in a signal and periodogram provides insight for the energy content at each frequency [25].

The analysis was performed on the raw EEG data and on the data after applying PCA and ICA transformations. This resulted in a richer feature set. Specifically:

- Raw EEG: PSD was computed using both FFT and periodogram methods.
- PCA-transformed EEG: PSD features were also extracted after reducing dimensionality using PCA, capturing key sources of variance in the data.
- ICA-transformed EEG: ICA was used to isolate independent components of the EEG signals and the PSD features were extracted from these components to reduce noise from artefacts such as eye blinks or muscle movements.

This comprehensive set of PSD features from multiple signal representations (raw, PCA, ICA) provided a robust characterisation of brain activity across different temporal and frequency scales.

7.2.2 Signal Statistical Features

In addition to the frequency domain features, statistical features from the time-domain EEG signals were extracted similar to what was done in [43]. These features provide insight into the distribution and variation of the signal over time. The following statistical measures were calculated specifically for each EEG channel: mean, standard deviation, skewness, kurtosis, and entropy.

- Mean: Represents the average signal amplitude.
- Standard deviation: Measures the variability of the signal, indicating how much the signal fluctuates from its mean.
- Skewness: Describes the asymmetry of the signal distribution, helping to capture the shape of the signal.
- Kurtosis: Reflects the “peakedness” of the signal distribution, offering insight into how extreme values influence the data.
- Entropy: Provides a measure of signal complexity, quantifying the unpredictability of the EEG signal.

These statistical features were calculated across all channels offering a further understanding of the temporal characteristics of the brain activity during the experiment.

7.2.3 Principal Component Analysis

Principal Component Analysis (PCA) was employed to reduce the dimensionality of the EEG data which transformed the raw signals into uncorrelated components that explain most of the variance in the original data. By applying PCA, we extracted a set of features from the transformed data that represent the key sources of variation in the brain signals. PSD features were computed for each principal component using both the FFT and periodogram methods. This allowed us to capture the dominant frequency patterns in the reduced-dimensionality EEG space, making the feature extraction process more efficient while retaining important information from the original signal.

7.2.4 Independent Component Analysis

Independent Component Analysis (ICA) is another dimensionality reduction technique. However, unlike PCA, ICA seeks to separate the EEG signals into statistically independent sources. This is particularly useful for removing artefacts such as eye blinks or muscle movements which can contaminate EEG data.

By applying ICA, the raw EEG signals were decomposed into independent components and the PSD features were then extracted from these components using both FFT and periodogram methods. This method allows for a more refined analysis of the brain signals as it isolates the neural activity from non-neural artefacts.

By combining both the raw EEG features with PCA and ICA-based features, the full feature vector provided a detailed and multi-dimensional representation of brain activity during the SA tests.

7.3 fNIRS Feature Extraction

7.3.1 Signal Statistical Features

Similar to EEG, statistical features were extracted from the fNIRS signals to capture the temporal characteristics of the hemodynamic responses. The raw fNIRS data consisted of oxygenated hemoglobin (O₂Hb) and deoxygenated hemoglobin (HHb) concentrations for each channel. The following statistical features were computed for both O₂Hb and HHb signals:

- **Mean:** The average concentration of hemoglobin across the signal.
- **Variance:** A measure of the signal's spread, indicating how much the concentration fluctuates.
- **Max values:** The peak concentration values of both O₂Hb and HHb, which may correspond to moments of peak brain activity.
- **Difference between O₂Hb and HHb:** Captures the balance between oxygenated and deoxygenated hemoglobin, which can indicate levels of brain activation.
- **Slope of the difference:** The rate of change in the difference between O₂Hb and HHb, offering insights into the dynamic nature of the hemodynamic response.

These statistical features are crucial for interpreting the physiological changes in brain activity during the SA tests.

7.3.2 Independent Component Analysis

ICA was also applied to the fNIRS data to separate the signals into independent components. This step is useful for isolating physiological noise (such as cardiac pulsation and respiration) from the neural signal. After applying ICA, statistical features were also extracted from the independent components to ensure that the most informative parts of the signal were used for further analysis.

The extracted features from the ICA components help in reducing noise while enhancing the sensitivity of the fNIRS signals to brain activity. By combining both statistical and ICA-based features, a comprehensive set of features was developed for the fNIRS data, improving the ability to classify the participant's cognitive state during the SA tests.

7.4 Feature Selection

After extracting a comprehensive set of features from the EEG and fNIRS signals, it was necessary to perform feature selection to reduce the dimensionality of the dataset. This step is important for improving model interpretability and avoiding overfitting. To achieve this, Recursive Feature Elimination (RFE) was employed as the primary method for feature selection.

7.4.1 Recursive Feature Elimination (RFE)

Recursive Feature Elimination (RFE) was chosen for feature selection due to its ability to iteratively rank and eliminate less important features based on their contribution to model performance. This process is well-suited for EEG and fNIRS data, where features often exhibit complex interactions that cannot be captured by simpler selection techniques. RFE was applied using a Random Forest classifier as the base estimator, which effectively handles non-linearities and interactions between features making it an ideal choice for this type of neurophysiological data.

The RFE method was configured to iteratively eliminate features and rank them based on their importance to the classifier. Instead of selecting a fixed number of top features, the model was trained and tested using a range of feature subsets. Specifically, the following feature set sizes were tested: 10, 20, 40, 80, 160, 200, 300 and up to the full set of available 408 features. This iterative approach enabled a thorough investigation of how model performance scales with the number of selected features. By varying the

number of features, it was possible to identify an optimal subset that balanced the trade-off between model complexity and classification performance.

The selected features from each iteration were used to train and test machine learning models, with the results compared across different feature set sizes. This approach ensured that the final model was trained on the most informative subset of features while maintaining high classification performance.

7.4.2 Consideration of Other Methods

In addition to RFE, two other feature selection techniques—Mutual Information (MI) and Pearson Correlation—were considered but ultimately not used for the final selection process.

Mutual Information measures the non-linear dependency between individual features and the target variable. However, MI only assesses the relationship between a single feature and the target, without accounting for interactions between features. Since EEG and fNIRS signals exhibit complex interdependencies across channels and modalities, MI was not ideal for capturing these interactions. On the other hand, RFE evaluates features in the context of their contribution to the model's performance providing a more holistic approach to feature selection.

Pearson Correlation measures the linear correlation between each feature and the target variable. While useful for identifying simple linear relationships, Pearson Correlation does not account for the non-linear dependencies present in neurophysiological data. Additionally, Pearson Correlation evaluates each feature independently which can lead to the exclusion of features that may only be informative in combination with others. Given the complexity of the EEG and fNIRS signals, Pearson Correlation was not deemed suitable for this study.

7.4.3 Top 10 Features

The top 10 features that were identified using the RFE method are presented in Table 4 below.

Table 4. *Top 10 features identified by RFE.*

Device	Channel Number	Method	Value
NIRS	1	ICA	Max O ₂ Hb
EEG	7	FFT ICA	Delta-band Power
EEG	6	FFT ICA	Delta-band Power
EEG	1	FFT ICA	Alpha-band Power
EEG	0	Periodogram	Delta-band Power
NIRS	5	ICA	Slope of the Difference
NIRS	0	ICA	Max O ₂ Hb
EEG	0	Periodogram ICA	Alpha-band Power
EEG	0	Periodogram ICA	Gamma-band Power
EEG	6	Periodogram ICA	Gamma-band Power

8 Model Development

To effectively classify situational awareness based on the EEG and fNIRS data, machine learning models were developed.

8.1 Model 1 – k-Nearest Neighbours (KNN)

8.1.1 Background

The k-Nearest Neighbours (KNN) algorithm is a non-parametric, instance-based learning method widely used for classification and regression problems [44]. KNN operates by retaining the entire training dataset during the learning phase and using it directly when making predictions during testing. Unlike parametric models such as logistic regression or support vector machines, KNN does not assume an underlying probability distribution of the data. Instead, it makes predictions by considering the majority class of the nearest neighbours to a given data point based on a defined distance metric. This method's simplicity and effectiveness for classification has popularised its use in various scientific and engineering disciplines.

The principle behind KNN is that it suggests that similar data points can be found near each other in the feature space [45]. Thus, for classification tasks, the algorithm assigns the class of a new sample based on the most frequent class among its nearest 'k' neighbours. The value of 'k' is a critical hyperparameter that controls the number of neighbours considered for the prediction. A small 'k' may lead to a model that is sensitive to noise in the data (overfitting), while a larger 'k' may smooth the decision boundary excessively (underfitting) [46].

In addition to choosing the appropriate 'k', KNN requires the selection of a distance metric to measure the "closeness" of points. Common distance metrics include Euclidean distance, Manhattan distance and Minkowski distance, each of which can affect the performance of the model depending on the nature of the data. Euclidean distance is most frequently used as it measures the straight-line distance between points in multidimensional space. Manhattan distance computes the sum of the absolute differences between coordinates making it more robust to outliers. Minkowski distance generalises both metrics and can be tuned using a parameter to balance between the two.

KNN is particularly useful in situations where the decision boundaries are non-linear and difficult to model with parametric approaches. However, the algorithm's performance degrades with increasing dimensionality (a phenomenon known as the "curse of dimensionality") because the distance between any two points becomes less informative in high-dimensional spaces. Therefore, dimensionality reduction techniques like Principal Component Analysis (PCA) are often used to mitigate this problem which improves both computational efficiency and model performance.

Figure 30 below shows an image to visually explain how the KNN algorithm works.

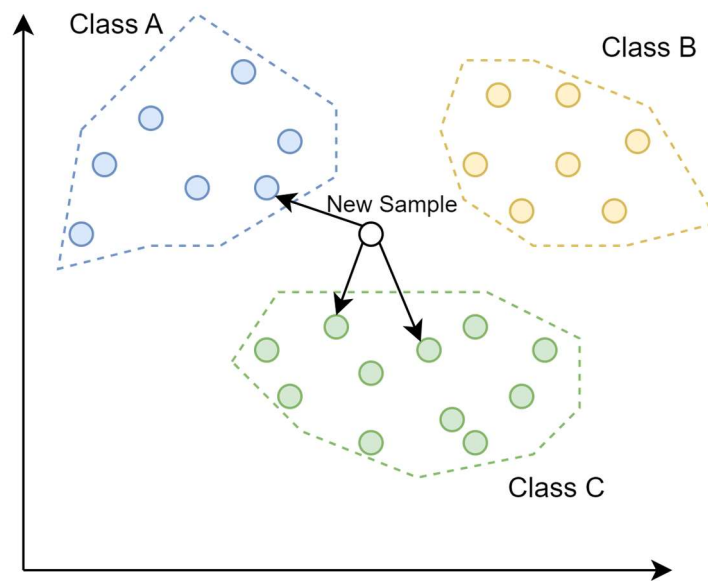


Fig. 30: Simple graphic to visualise the KNN algorithm ($k=3$).

8.1.2 Implementation

In this thesis, KNN was implemented to classify situational awareness levels based on features extracted from EEG and fNIRS signals. Due to the high dimensionality of the extracted features, Principal Component Analysis (PCA) was first applied to reduce the dimensionality of the dataset ensuring that the KNN algorithm could operate more efficiently. Specifically, PCA was used to reduce the data to 3 principal components which captured most of the variance in the data while eliminating irrelevant information. This step was essential for minimising the "curse of dimensionality" and enhancing model performance.

Following the dimensionality reduction step, the KNN algorithm was trained and tested on several subsets of features. The grid search technique was used to optimise key hyperparameters, including the number of neighbours ('k') and the distance metric. The optimal 'k' value was determined by evaluating model performance across a range of values (e.g., $k=1$ to $k=10$). Additionally, three distance metrics—Euclidean, Manhattan, and Minkowski—were tested to identify the most suitable metric for this dataset.

The model was implemented using Python's scikit-learn library and cross-validation was applied to mitigate overfitting and provide more reliable estimates of the model's generalisation performance. Grid search and cross-validation were integral in ensuring the final model was not only accurate but also generalised well to unseen data.

8.2 Model 2 – XGBoost (Boosted Trees)

8.2.1 Background

Boosted Trees is a powerful machine learning technique that builds an ensemble of decision trees to improve model accuracy and reduce bias. One of the most popular implementations of this technique is XGBoost (eXtreme Gradient Boosting) which has gained significant popularity for its high performance and flexibility in various machine learning competitions and applications [47]. The fundamental idea behind XGBoost is to combine multiple weak learners (decision trees) into a strong learner by iteratively training each new tree to correct the errors made by the previous trees. Unlike traditional decision trees, which are prone to overfitting, Boosted Trees use an additive model to minimise a loss function making it more robust and effective for complex datasets [47].

XGBoost is built on the principles of gradient boosting where the algorithm iteratively optimises a loss function by adding new trees that predict the residuals (errors) from the previous iteration. The advantage of XGBoost over traditional gradient boosting lies in its efficiency and scalability. It includes advanced features such as:

- Regularisation: To control overfitting, XGBoost introduces L1 and L2 regularisation terms, making it more robust on noisy data.
- Sparsity-aware learning: XGBoost is optimised for sparse data and can handle missing values efficiently.

- **Parallelised tree construction:** The algorithm supports parallel processing, which speeds up the model training process.
- **Automatic handling of missing data:** Unlike some algorithms that require preprocessing to fill missing values, XGBoost can automatically decide which way to split missing data.

Due to its superior handling of large, high-dimensional datasets, XGBoost has been widely adopted in applications such as time-series forecasting and recommendation systems.

8.2.2 Implementation

In this thesis, XGBoost was used to classify situational awareness levels based on EEG and fNIRS features. Given the high-dimensional nature of the dataset, XGBoost's ability to handle complex, non-linear relationships made it a strong candidate for the classification task. The implementation followed a systematic approach to optimise the model's performance using both feature selection and hyperparameter tuning.

Hyperparameter tuning was carried out using a grid search technique. The hyperparameters tuned in this process included:

- **Learning rate:** Controls the contribution of each tree to the final model. Lower values generally improve performance but require more iterations.
- **Number of estimators:** Specifies the number of trees in the model. Too few trees can lead to underfitting, while too many trees can cause overfitting.
- **Max depth:** Controls the depth of each tree. Deeper trees can model more complex patterns but are prone to overfitting.

The grid search was performed over a predefined range of values for these hyperparameters with cross-validation used to evaluate each combination. This process ensured that the model was optimised for accuracy without overfitting to the training data.

XGBoost's ability to handle non-linear relationships in the data was particularly important for the classification of situational awareness based on neurophysiological data, where the relationship between features and outcomes may not be easily captured by simpler models. Additionally, the use of regularisation (via L1 and L2 penalties) in

XGBoost helped mitigate overfitting allowing the model to generalise better to unseen data. The model was implemented using the XGBoost Python library.

8.3 Model 3 – Multi-layer Perceptron

8.3.1 Background

Multi-Layer Perceptron (MLP) is a type of feedforward artificial neural network widely used for solving both regression and classification tasks. It is particularly effective in capturing complex non-linear relationships between input features and output labels due to its multiple layers of neurons (units) and non-linear activation functions [48]. MLPs consist of three main types of layers: an input layer, one or more hidden layers and an output layer. Each layer is fully connected to the subsequent layer with each connection weighted and adjusted through a backpropagation algorithm during training [48].

The key strength of MLPs lies in their ability to learn hierarchical feature representations enabling them to model intricate patterns in the data that might not be captured by linear models or simpler algorithms like KNN [49]. By using non-linear activation functions such as ReLU (Rectified Linear Unit) and the addition of multiple hidden layers, MLPs excel in tasks where the decision boundary between classes is non-linear.

However, the challenge lies in the sensitivity of MLPs to hyperparameters such as learning rate, the number of hidden units, and regularisation techniques which can heavily influence their generalisation performance [48]. To prevent overfitting, regularisation strategies such as dropout are typically employed [50].

8.3.2 Implementation

For this thesis, the Multi-Layer Perceptron (MLP) was implemented using the TensorFlow and Keras libraries. The architecture consists of an input layer that receives the EEG and fNIRS features, followed by four hidden layers with 128, 64, 32, and 16 neurons respectively. The activation function used for all hidden layers is ReLU which is a standard choice for deep learning models due to its ability to introduce non-linearity without suffering from the vanishing gradient problem. Dropout layers with a rate of 0.3 are applied after each hidden layer to mitigate overfitting by randomly disabling a fraction of the neurons during each training iteration.

The output layer consists of a single neuron with a sigmoid activation function which is appropriate for binary classification tasks. Since this study is performing binary classification, the output is either 0 or 1.

Figure 31 below illustrates the architecture of the MLP used in this thesis.

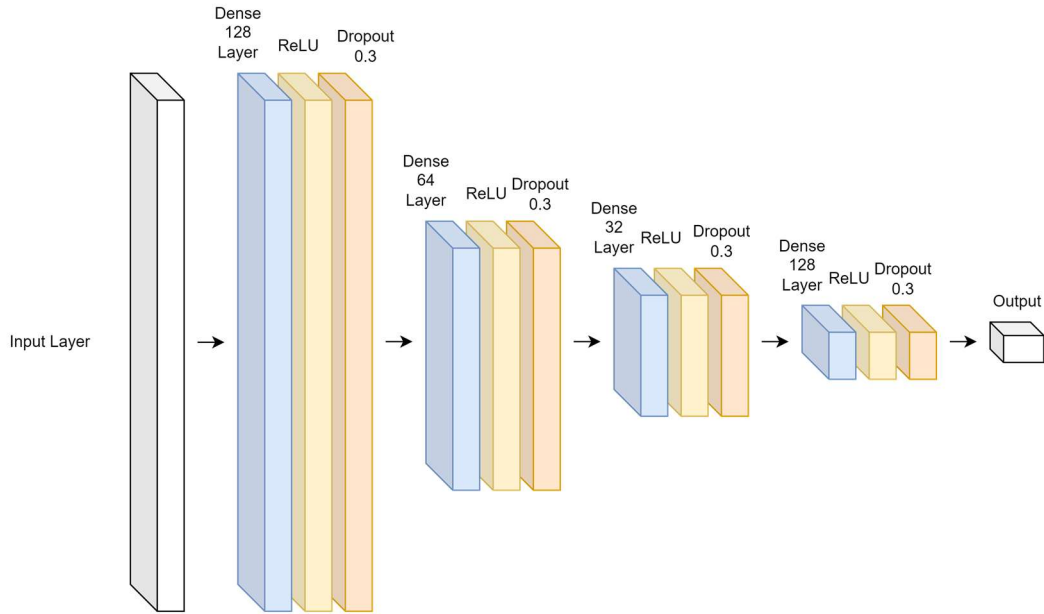


Fig. 31 Architecture of MLP implemented in this thesis.

Key implementation details are as follows:

- **Learning Rate:** The Adam optimiser was employed with a learning rate of 0.001, which is commonly used for its efficiency in training deep networks and adaptively adjusting the learning rate for each parameter.
- **Batch Size and Epochs:** The model was trained with a batch size of 32 and for a maximum of 150 epochs. Early stopping was used with a patience level of 5, ensuring that training halts if validation performance does not improve over 5 consecutive epochs which helps prevent overfitting.
- **Training Procedure:** Training was carried out with a validation split of 20%, meaning that 20% of the training data was reserved for validation during training. This ensured that the model's performance on unseen data was continuously monitored.

- Cross-validation (using K-fold validation) was not explicitly used for this model, though early stopping with validation split achieves a similar effect of preventing overfitting by monitoring performance on held-out data during training.

8.4 Model 4 – Transformer

8.4.1 Background

The Transformer model, introduced by Vaswani et al. in 2017, has become a foundational architecture in natural language processing (NLP) and time-series classification tasks [51]. Unlike traditional recurrent neural networks (RNNs) or long short-term memory networks (LSTMs), the Transformer architecture does not rely on sequential data processing. Instead, it leverages self-attention mechanisms to process input data in parallel which allows for faster training and greater flexibility in handling long-range dependencies in time-series data [52].

The self-attention mechanism computes the relationships between all elements in the input sequence at once which makes it highly effective for understanding the dependencies between various time steps in time-series signals [51]. By using multiple heads in the attention layers, Transformers can capture different representations of the data simultaneously which makes them more effective for complex classification tasks [53].

In recent years, transformers have been applied beyond NLP tasks finding utility in domains like computer vision and time-series data classification. For time-series classification, a Transformer can learn patterns across multiple time steps which is critical for identifying situational awareness states from neurophysiological data.

8.4.2 Implementation

For this thesis, a transformer-based model was implemented to classify EEG and fNIRS features. The Transformer model used in this study consisted of several key components:

- **Multi-head Self-Attention:** The model employs multi-head attention layers to capture dependencies across the input sequence simultaneously. Each head processes differently, helping the model understand various aspects of the features.

- **Feed-forward Layers:** After the attention mechanism, the outputs are passed through a series of feed-forward layers that help in transforming the attention outputs into meaningful features for classification.
- **Global Average Pooling:** The model reduces the feature dimensions using global average pooling, which helps summarise the temporal information across the sequence.

The model was trained with a K-Fold cross-validation strategy, splitting the data into five folds to ensure robustness. During each fold, the training data was split into validation sets and the model was optimised using early stopping and learning rate reduction based on the validation performance.

Key Parameters:

- **Input Shape:** The EEG and fNIRS features were reshaped into a 3D tensor of shape (num_samples, 1, num_features) to be compatible with the Conv1D layers used in the Transformer model.
- **Attention Mechanism:** Each attention layer used 256 dimensions (head size) and four heads for multi-head attention. This configuration allowed the model to learn dependencies from various perspectives.
- **Dropout:** To prevent overfitting, dropout layers with a probability of 0.4 were included after the attention layers and feed-forward layers.

The final output of the model was a classification layer with a softmax activation function to predict the probability of the situational awareness levels.

Figure 32 below illustrates the architecture of the Transformer model used in this thesis.

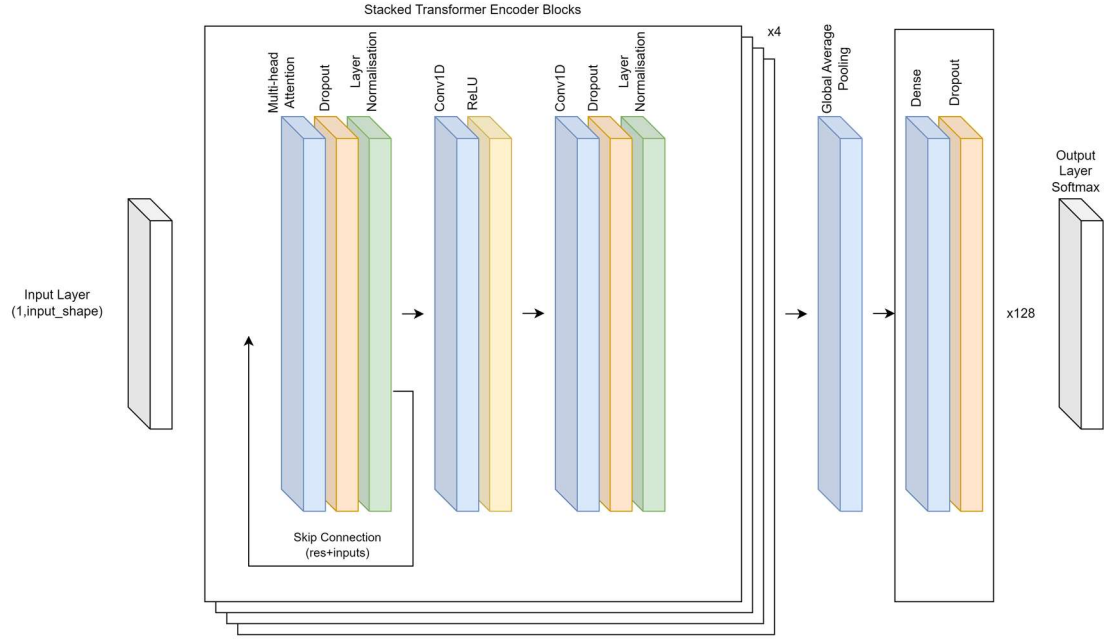


Fig. 32 *Architecture of Transformer model implemented in this thesis.*

Model Training:

- **Optimiser:** The model was optimised using the Adam optimiser with an initial learning rate of $1e-4$. Learning rate reduction and early stopping were applied to halt training when the validation loss stopped improving.
- **Batch Size:** A batch size of 64 was used to process the data which helped balance computational efficiency with model convergence.

9 Results and Discussion

9.1 Model Performance

In this section, the performance results will be presented of the four models – k-Nearest Neighbours (KNN), XGBoost, Multi-Layer Perceptron (MLP) and Transformer – across varying feature set sizes. The feature sets ranged from 10 to 408 features and the feature set size that each of the models performed the best will be shown. For each model, the key performance metrics such as accuracy, precision, recall and F1-score are summarised below in Table 5.

Table 5. Model performance results.

	KNN	XGBoost	MLP	Transformer
Test Accuracy	0.544	0.588	0.559	0.603
Precision (Low SA)	0.55	0.88	0.60	0.68
Precision (High SA)	0.54	0.55	0.54	0.57
Recall	0.54	0.59	0.56	0.60
F1-Score	0.54	0.52	0.54	0.58
No. Features	200	200	200	10

9.2 Discussion

The results indicate key insights into how different machine learning models handle the task of classifying situational awareness using features derived from EEG and fNIRS signals. The four models evaluated show varying degrees of effectiveness, depending on the number of features used in the overall model architecture.

9.2.1 Overall Model Performance

Among the models, the Transformer demonstrated the highest test accuracy of 60.3%, outperforming the other models. XGBoost followed with 58.8%, while MLP and KNN showed test accuracies of 55.9% and 54.4% respectively. This demonstrates that the Transformer model, with its self-attention mechanism, is capable of capturing intricate

patterns within the data more effectively than traditional models such as KNN and MLP.

Interestingly, the Transformer model reached its peak performance with only 10 features while the other models performed best with 200 features. This suggests that the self-attention mechanism in the Transformer is particularly efficient at extracting meaningful information from a condensed feature set, even when the input dimensionality is reduced. This is a marked contrast to traditional models like KNN and MLP which tend to perform better with larger feature sets, likely because they rely more on high-dimensional spaces to draw decision boundaries.

One notable aspect of this implementation is that the Transformer model was fed pre-extracted features with the shape (1, num_features) rather than raw time-series data. The original Transformer architecture was designed for sequential data (e.g., time-series or language data) where the model processes tokens in a sequence while learning temporal dependencies. Despite this, the model still performed well in this context, demonstrating its versatility. However, using extracted features may have limited the Transformer's potential as the model's core strength lies in its ability to process and learn temporal relationships within the raw time-series data. If raw EEG and fNIRS signals had been used, the Transformer might have been able to leverage its self-attention mechanism more effectively to capture subtle temporal dynamics, potentially leading to higher accuracy.

While the Transformer outperformed other models, it is important to acknowledge its higher computational complexity compared to simpler models like KNN or XGBoost. For situations where computational resources are limited or where training time is a concern, XGBoost could still be a viable option given its competitive performance especially for the low SA class despite its lower recall for high SA.

9.2.2 Performance Across Metrics

While accuracy is a key measure of the model's ability to predict the correct situational awareness level, the precision, recall, and F1-score provide further insight into the models' effectiveness in identifying low and high SA levels.

- Precision: XGBoost achieved the highest precision for the "Low SA" class (0.88) meaning it was highly effective at correctly identifying instances of low situational

awareness. However, its precision for the "High SA" class was notably lower (0.55) indicating that while XGBoost can identify low SA accurately, it struggles to correctly identify high SA. The Transformer showed more balanced precision between the two classes with 0.68 for low SA and 0.57 for high SA indicating better generalisation across the different situational awareness levels.

- **Recall:** All models performed similarly in terms of recall with Transformer slightly outperforming others at 0.60. This means that while each model was effective at identifying a substantial portion of the actual positive cases, the Transformer's ability to detect true positives was marginally better than KNN, MLP, and XGBoost.
- **F1-Score:** The F1-score, which balances precision and recall, showed that the Transformer once again had the best overall performance (0.58). The relatively balanced F1-score indicates that the Transformer achieved a better trade-off between precision and recall, particularly compared to XGBoost which had a higher precision but lower recall resulting in a less balanced F1-score.

9.2.3 Comparison to Literature

The study's results can be compared to the findings in Kästle et al. (2021), which focused on classifying situational awareness using EEG signals and machine learning models [25]. In that study, Gaussian mixture models were used to derive thresholds for classifying situational awareness which worked effectively due to their large and varied participant pool (32 participants). However, this study only included data from a single participant which likely limited the applicability of the thresholds derived from [25]. When these generalised thresholds were applied to this more limited dataset, the classification performance suffered potentially contributing to the lower overall accuracies across all models. For future work, recalculating participant-specific thresholds or dynamically adjusting thresholds could improve model accuracy.

Additionally, [25] used carefully controlled EEG impedance to ensure high-quality signals whereas in this study, higher than optimal EEG impedance likely introduced noise into the data which further contributed to the lower performance of the models. Maintaining lower EEG impedance (ideally below 20 kOhms) would reduce noise and improve signal quality, resulting in better classification performance particularly for models sensitive to input data quality, such as the Transformer.

Despite these challenges, it has been successfully demonstrated that there is some correlation between EEG and fNIRS signals and situational awareness levels. While the accuracy and F1-scores were lower than expected, this result shows promise especially considering the small sample size and data quality issues that were faced.

9.2.4 Lessons from the Use of Extracted Features

Although the Transformer model achieved the highest accuracy, it is important to note that it was fed pre-extracted features rather than raw time-series signals. This may have limited the model's ability to fully leverage its strengths in learning temporal dependencies between data points. The Transformer architecture is inherently designed for sequential data processing, where self-attention can capture long-range dependencies across time steps. By using pre-extracted features, the temporal structure of the EEG and fNIRS data was likely lost, and the model was constrained to learn only from static feature representations.

Despite this limitation, the Transformer still managed to perform well, suggesting that the extracted features captured meaningful aspects of the neurophysiological signals. However, for future experiments, it would be worthwhile to test the model with raw time-series data as input to allow the Transformer to fully utilise its temporal modelling capabilities. This might result in even better classification performance as the model could capture both the temporal and feature relationships in the data.

10 Conclusions from the work

The aim of this thesis was to explore the classification of situational awareness (SA) levels using a combination of EEG and fNIRS data, with the goal of developing and testing machine learning models capable of identifying SA states in real time. To achieve this, four different models—k-Nearest Neighbours (KNN), XGBoost, Multi-Layer Perceptron (MLP), and Transformer—were evaluated on their ability to classify SA based on features extracted from neurophysiological signals.

The thesis began by detailing the physiological underpinnings of situational awareness, EEG, and fNIRS technologies, followed by a thorough review of relevant literature. EEG and fNIRS were chosen because of their complementary properties; EEG provides high temporal resolution while fNIRS provides better spatial resolution. Together, they offer a more comprehensive view of brain activity during tasks requiring high levels of awareness.

The methodology involved designing a custom experiment to collect data from a single participant while completing a situational awareness test. Data pre-processing techniques were applied to clean the EEG and fNIRS signals, including filtering, Independent Component Analysis (ICA) and Principal Component Analysis (PCA). The cleaned data was then used to extract features, which served as the input for the machine learning models.

Recursive Feature Elimination (RFE) was employed to select the most relevant features and different models were trained using various feature sets (ranging from 10 to 408 features). Performance was evaluated using accuracy, precision, recall, and F1-score.

The results demonstrated that the Transformer model outperformed the other models achieving the highest accuracy (60.3%) using only 10 features. This suggests that the self-attention mechanism of the Transformer model effectively captures complex relationships within the feature set, even when the input dimensionality is reduced. XGBoost also performed well, particularly in identifying low SA levels but its performance was more variable when identifying high SA states.

Challenges were identified in terms of thresholding and signal quality. The thresholds for SA classification, derived from previous literature [25], may not have been optimal for the single-participant dataset used in this study, leading to lower overall accuracies. Furthermore, higher than ideal EEG impedance likely introduced noise into the signals which further impacted model performance.

Another key point to consider is the implementation of the Transformer model in this study. Unlike traditional approaches that feed raw time-series signals into the model, this thesis used extracted features as inputs. While the Transformer still performed well, it is worth noting that Transformers are typically designed to work with sequential data and applying them to non-sequential, feature-based data may not fully leverage their architecture's strengths.

Despite these challenges, this thesis successfully demonstrates the potential of using EEG and fNIRS data combined with advanced machine learning techniques to classify situational awareness. The study also highlights the versatility of Transformer models which showed robust performance even when provided with pre-extracted features instead of raw time-series signals.

Future work should focus on expanding the dataset to include multiple participants, which would enable the development of more generalisable models. Additionally, further improvements in EEG signal quality, individualised thresholding methods and the exploration of raw time-series input into the Transformer could enhance model accuracy and robustness.

In conclusion, this thesis contributes to the growing body of work on situational awareness classification using neurophysiological signals. The combination of EEG, fNIRS, and advanced machine learning techniques offers a promising approach to real-time monitoring of awareness in high-risk environments providing valuable insights for future research and practical applications.

11 Future work

While this thesis successfully demonstrated the use of EEG and fNIRS signals for situational awareness (SA) classification, several challenges and limitations emerged that require attention in future work. Addressing these limitations would improve the accuracy, generalisability and practical applicability of the models developed in this study. The following subsections outline the key areas for improvement and future directions.

11.1 Data Collection and Generalisability

One of the primary limitations of this study was the small dataset—data was collected from only one participant. While this allowed for proof of concept and preliminary testing of the models, it limited the generalisability of the results. Future work should focus on expanding the participant pool to include a larger and more diverse sample of individuals. This would not only increase the robustness of the models but also allow for better thresholding methods that can accommodate inter-participant variability in EEG and fNIRS signals.

Additionally, including multiple participants would facilitate the development of personalised thresholding methods for situational awareness classification. The thresholds derived from [25] were applied to a single participant in this study, which may have introduced inaccuracies due to differences in brainwave patterns and responses to stimuli across individuals. Implementing adaptive or personalised thresholds would likely improve the classification performance for each participant.

11.2 EEG Signal Quality and Impedance

The higher than ideal EEG impedance encountered during data collection may have introduced noise into the signals, adversely affecting model performance. In future experiments, greater attention should be given to managing impedance levels ideally keeping them below 20 kOhms as recommended in medical settings. This can be achieved through better skin preparation, improved electrode placement and possibly using more advanced dry electrode systems or gel-based electrodes to ensure consistent signal quality.

Moreover, implementing real-time impedance monitoring during data collection would allow for immediate adjustments, minimising noise and improving the overall quality of the EEG data. This would directly benefit model accuracy, particularly for models like the Transformer, which are sensitive to input quality.

11.3 Model Improvement: Use of Raw Time-Series Data

While the models in this thesis were trained using extracted features, future work should explore the possibility of using the pre-processed time-series EEG and fNIRS signals as the input to the transformer. The Transformer architecture is specifically designed to handle sequential data and can leverage temporal relationships across time steps. By using time-series signals, the model could potentially uncover complex temporal patterns that were lost during feature extraction. This may lead to higher classification accuracy and better generalisation.

For this, it would be important to modify the current model implementation to handle time-series data efficiently, such as adjusting the input dimensionality to capture relevant temporal information. Additionally, preprocessing techniques to handle baseline drift and artifact rejection should be fine-tuned to ensure the raw signals are of high quality before feeding them into the model.

11.4 Thresholding and SA Test Design

The current SA test design utilised pre-determined thresholds for categorising situational awareness levels which were derived from previous literature. As mentioned earlier, these thresholds may not have been suitable for the single participant involved in this study. To improve this aspect of the project, future work should focus on dynamic thresholding techniques that adapt to the participant's performance and physiological responses over time. For example, Gaussian Mixture Models (GMMs) or k-means clustering could be explored.

Furthermore, enhancing the complexity and variety of tasks within the SA test could also improve the robustness of the results. A more complex test would better simulate real-world conditions where SA is critical, such as in military or industrial settings. Incorporating real-world tasks that require higher cognitive engagement could yield more representative data, making the models more applicable to practical situations.

11.5 Cross-Model Comparison and Hybrid Approaches

While the Transformer model outperformed other models in this study, future research could explore hybrid model architectures that combine the strengths of different models. For example, a combination of XGBoost's ability to handle structured data and the Transformer's ability to capture sequential relationships might yield better performance. Cross-model comparisons should continue to focus on accuracy, precision, recall, and computational efficiency, particularly when applied to larger datasets.

In summary, future work on this project should focus on improving data quality, expanding the participant pool and exploring more advanced modelling techniques, including the use of raw time-series data. By addressing these limitations and incorporating personalised approaches, the classification of situational awareness using EEG and fNIRS signals can be significantly improved. These advancements will also ensure that the models developed can be applied to real-world scenarios, such as monitoring workers in high-risk environments, enhancing the practical applicability of this research.

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13 Appendices

13.1 Appendix A: Multi-Input CNN with windowing technique

Leelasiri, et al. [54] use EEG and fNIRS to classify voluntary left-hand and right-hand movements by participants. In [54], a multi-input CNN was implemented to handle the 2 different frequency signals from EEG and fNIRS. The highest accuracy achieved was approximately 90% [54]. The architecture proposed in this literature attempts to accommodate for the differing frequencies without affecting the characteristics of the signals.

Figure 33 below shows a diagram of the model architecture that was implemented by Leelasiri, et al. [54].

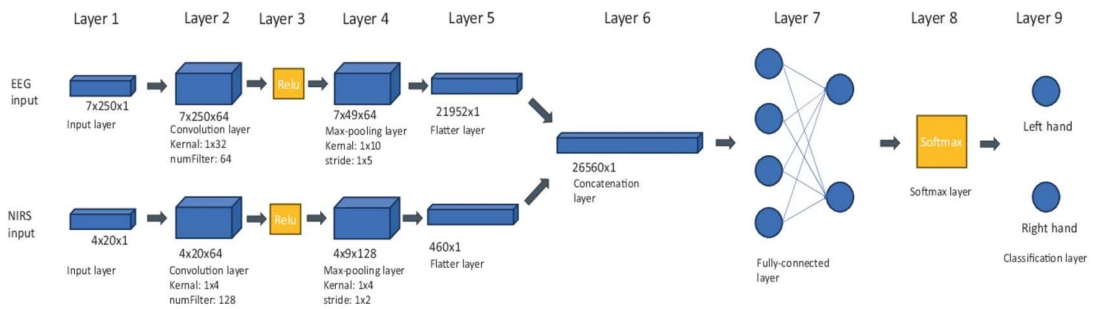


Fig. 33 Structure of the multi-cnn deep neural network [54].

This architecture has been implemented for use in this thesis. As can be seen in Figure 33, there are 2 separate branches for the 2 different inputs which contain a convolutional layer, max-pooling layer and then flattening layer. The branches are then concatenated and passed into a fully connected layer. The input dimensions are not the same as shown in Figure 33 and have been modified because the signal duration is varied for the data obtained in this thesis. A function for creating the architecture was made using TensorFlow and can be seen below.

```
def model(self):  
    # EEG input  
    eeg_input = layers.Input(shape=self.eeg_input_dims, name="eeg_input")  
  
    # EEG branch  
    h_eeg = layers.Conv2D(  
        filters=64, kernel_size=(32, 1), activation="relu"  
    )(eeg_input)  
    h_eeg = layers.MaxPooling2D(  
        pool_size=(10, 1), strides=(5, 1)  
    )(h_eeg)  
    h_eeg = layers.Flatten()(h_eeg)
```

```

# NIRS input
nirs_input = layers.Input(shape=self.nirs_input_dims, name="nirs_input")

# NIRS branch
h_nirs = layers.Conv2D(
    filters=128, kernel_size=(4, 1), activation="relu"
)(nirs_input)
h_nirs = layers.MaxPooling2D(
    pool_size=(4, 1), strides=(2, 1)
)(h_nirs)
h_nirs = layers.Flatten()(h_nirs)

# Concatenate both branches
concatenated = layers.Concatenate()([h_eeg, h_nirs])

# Fully connected layer
fc = layers.Dense(2, activation="relu")(concatenated)

# Output layer with softmax for classification
output = layers.Dense(2, activation="softmax", name="output")(fc)

# Create model
model = models.Model(inputs=[eeg_input, nirs_input], outputs=output)

return model

```

This model input data for this model was 1-second windows of the raw fNIRS and EEG signals. The model did not provide good results at all and from this point, it was realised that feature extraction is an important step for brain activity data. This is especially true for attempting to predict situational awareness as the raw activity will contain lots of noise and artefacts that will affect the result.

13.2 Appendix B: Data Pre-Processing Code

The code for pre-processing the EEG data is shown below.

```

def EEGData(path_to_data_folder):
    # Band Pass Filter function
    def bandpass_filter(data, lowcut, highcut, fs, order=5):
        nyquist = 0.5 * fs
        low = lowcut / nyquist
        high = highcut / nyquist
        b, a = butter(order, [low, high], btype='band')
        return filtfilt(b, a, data, axis=0)

    # Notch filter function to remove powerline noise
    def notch_filter(data, freq, fs, quality_factor=30):
        nyquist = 0.5 * fs
        freq = freq / nyquist
        b, a = butter(2, [freq - 0.01, freq + 0.01], btype='bandstop')
        return filtfilt(b, a, data, axis=0)

```

```

# Function to replace outliers with median values
def replace_outliers_with_median(data, threshold=4):
    # Calculate mean and standard deviation along each column
    mean = np.mean(data, axis=0)
    std_dev = np.std(data, axis=0)

    # Calculate lower and upper bounds for outliers
    lower_bound = mean - threshold * std_dev
    upper_bound = mean + threshold * std_dev

    # Identify outliers (values outside the lower and upper bounds)
    outliers = (data < lower_bound) | (data > upper_bound)

    # Calculate the median for each column
    median = np.median(data, axis=0)

    # Replace outliers with the median of the respective column
    for col in range(data.shape[1]): # Iterate through each column
        data[outliers[:, col], col] = median[col]

    return data

# Load and pre-process EEG data
EEG_df = readAllEEGData(path_to_data_folder)
# Keep only the useful columns
EEG_df = EEG_df.drop(
    columns=EEG_df.columns.difference([
        "EXG Channel 0",
        "EXG Channel 1",
        "EXG Channel 2",
        "EXG Channel 3",
        "EXG Channel 4",
        "EXG Channel 5",
        "EXG Channel 6",
        "EXG Channel 7",
        "Timestamp (Formatted)"
    ]), axis=1
)

# Extract EEG signals
eeg_signals = EEG_df.iloc[:, :-1].values

# Parameters for filtering
fs = 250 # Sampling frequency (Hz)
lowcut = 0.5 # Low cutoff frequency
highcut = 45 # High cutoff frequency

```

```

# Replace the EEG signals in the DataFrame
EEG_df.iloc[:, :-1] = eeg_signals

# Apply notch filter to remove power line interference at 50Hz
filtered_eeg = notch_filter(eeg_signals, freq=50, fs=fs)

# Apply bandpass filter (e.g., 0.5 Hz to 50 Hz) - this removes DC component as
well
filtered_eeg = bandpass_filter(filtered_eeg, lowcut, highcut, fs)

# Remove outliers
filtered_eeg = replace_outliers_with_median(filtered_eeg, threshold=4)

# Replace the EEG signals in the DataFrame
EEG_df.iloc[:, :-1] = filtered_eeg

return EEG_df

```

The code for pre-processing the fNIRS data is shown below.

```

def NIRSDData(path_to_data_folder):
    # Apply low-pass filter to remove high-frequency noise (e.g., >0.5Hz)
    def lowpass_filter(data, cutoff, fs, order=5):
        nyquist = 0.5 * fs
        cutoff = cutoff / nyquist
        b, a = butter(order, cutoff, btype='low')
        return filtfilt(b, a, data, axis=0)

    # Fixing outliers
    def replace_outliers_with_median(data, threshold=4):
        # Calculate mean and standard deviation along each column
        mean = np.mean(data, axis=0)
        std_dev = np.std(data, axis=0)

        # Calculate lower and upper bounds for outliers
        lower_bound = mean - threshold * std_dev
        upper_bound = mean + threshold * std_dev

        # Identify outliers (values outside the lower and upper bounds)
        outliers = (data < lower_bound) | (data > upper_bound)

        # Calculate the median for each column
        median = np.median(data, axis=0)

        # Replace outliers with the median of the respective column
        for col in range(data.shape[1]): # Iterate through each column
            data[outliers[:, col], col] = median[col]

```

```

    return data

# Load and consolidate all NIRS data from the specified folder
NIRS_df = readAllNIRSData(path_to_data_folder)

# Extract NIRS signals (assuming NIRS_df has 'HbO' and 'HbR' columns)
nirs_signals = NIRS_df.iloc[:, :-1].values

# Parameters for filtering
fs = 50 # Sampling frequency (Hz)
cutoff = 0.7

# Apply bandpass filter to remove high-frequency and low-frequency noise
filtered_nirs = lowpass_filter(nirs_signals, cutoff, fs)

# Remove outliers
filtered_nirs = replace_outliers_with_median(filtered_nirs)

# Replace the NIRS signals in the DataFrame
NIRS_df.iloc[:, :-1] = filtered_nirs

return NIRS_df

```